



**Canaan**  
迦南凯鑫隆

## 说明书

### NJP3500全自动胶囊充填机

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# 一、前言

## Preface

随着制药机械日益更新,如何提高制药设备的自动化程度和生产效率已成为制药企业迫切要求。为适应大、中、小药厂的生产与发展需要,我厂在原生产 NJP 系列充填机基础上,开发成功 NJP-3500 型结构封闭式的全自动硬胶囊充填机。本厂生产的 NJP-3500 型全自动胶囊充填机是在以往的基础上取其精华,锐意创新,进行大量改进而成。该机型在技术设备、工位传动、清洁机械、点动控制系统等方面均作了创新性的改进设计,多项技术指标达到了国际同类产品领先水平,它是集机、电、气一体化高新技术产品,是目前国内最理想,最完美的生产胶囊剂药品的充填设备。

With the continuous updating of pharmaceutical machinery, how to improve the automation level and production efficiency of pharmaceutical equipment has become an urgent requirement for pharmaceutical enterprises. In order to meet the production and development needs of large, medium, and small pharmaceutical factories, our factory has successfully developed the NJP-3500 fully automatic hard capsule filling machine based on the original NJP series filling machine. The NJP-3500 automatic capsule filling machine produced by our factory is based on the essence of the past, boldly innovating and making a lot of improvements. This model has made innovative improvements in technical equipment, workstation transmission, cleaning machinery, and

incremental control systems. Many technical indicators have reached the leading level of similar products internationally. It is a high-tech product integrating machinery, electricity, and gas, and is currently the most ideal and perfect filling equipment for capsule drugs production in China.

## **二、机器的基本信息**

### **II. Basic information of the machine**

#### **1、质保期和责任**

##### **1. Warranty Period and Responsibility**

质保期和责任根据在合同中规定的条款和条件执行。

The warranty period and responsibility shall be implemented in accordance with the terms and conditions specified in the contract.

##### **1.1 软件修改**

##### **1.1 Software Modification**

如没有得到迦南凯鑫隆公司的授权和允许进行软件修改,质保义务将终止。

If the authorization and permission from Canaan Kexin Company are not obtained for software modification, the warranty obligation will be terminated.

##### **1.2 说明图示和图纸**

##### **1.2 Explanation of Diagrams and Drawings**

操作手册中的图示和图纸仅用于一般阐释说明。实际设备与图示可能存在差异。手册中的图示和图纸所标明的尺寸不具有约束性。

Diagrams and drawings in the operation manual are for general

explanatory purposes only. There may be differences between the actual equipment and the diagrams. The dimensions indicated in the diagrams and drawings in the manual are not binding.

### **1.3 所有权**

#### **1.3 Ownership**

迦南凯鑫隆公司保留其对所有图纸、软件、其他文件及所有相关专利申请的权利，保留对其销毁、复制和传播的权利。

Canan Kaixinlong Company reserves the rights to all drawings, software, other documents, and all related patent applications, and reserves the rights to their destruction, reproduction, and dissemination.

### **1.4 储存**

#### **1.4 Storage**

任何由于储存不当造成的腐蚀损坏，迦南凯鑫隆公司不承担任何保修义务。

Canan Kaixinlong Company shall not be liable for any warranty obligations due to corrosion damage caused by improper storage.

### **1.5 运输**

#### **1.5 Transportation**

如有合适的迦南凯鑫隆人员在场，可向其询问运输事宜。但是，对于运输过程中的任何损害，迦南凯鑫隆人员不承担任何责任。

If there are suitable Canan Kaixinlong personnel present, you can inquire with them about transportation matters. However, Canan Kaixinlong

personnel shall not be liable for any damages during the transportation process.

## 2、拟定用途

### Intended Use

该机器仅能在订单确认的范围进行工作。任何超出订单范围的用途，如生产未在订单确认范围内的产品，均视为非正常使用。

The machine can only operate within the scope confirmed in the order. Any use beyond the scope of the order, such as producing products not included in the order confirmation, shall be considered abnormal use.

迦南凯鑫隆公司对于非正常使用而产生的机器损坏不负有任何责任。风险由机器所有者/使用者承担。

Canan Kaixinlong Company shall not be responsible for any machine damage caused by abnormal use. The risk shall be borne by the machine owner/user.

#### 2.1 禁止操作模式

##### 2.1 Prohibited Operating Modes

A.禁止使用未列入订单允许范围的电源、产品、包装材料、包装介质等。

A. Prohibited use of power sources, products, packaging materials, packaging media, etc., not included in the allowed scope of the order.

B.禁止使用属于危险物或者任何对操作者健康有害的电源、产品、包装材料、包装介质等。

B. Prohibit the use of dangerous substances or any power sources, products, packaging materials, packaging media that are harmful to the

operator's health.

C.禁止使用擅自更改过的机器装置。

C. Prohibit the use of machines with unauthorized modifications.

## 2.2 制药机械相关说明

### 2.2 Pharmaceutical Machinery Related Instructions

① 为保护操作人员免受生产时的危险，如需增加合同允许的机器附件作为保护装置，此类保护装置应安装在机器所需位置，且必须证明其有足够的可操作性。

To protect operators from hazards during production, if additional machine accessories are needed as protective devices beyond the contract allowance, such protective devices should be installed in the required positions of the machine and must demonstrate sufficient operability.

② 机器所有者/操作者须采取足够的预警措施，保护全体职员远离生产危害。应特别注意以下情况：要在机器上实施正常标准操作以外的作业（如清洁、废物处理、修理）。

Machine owners/operators must take sufficient precautionary measures to keep all staff away from production hazards. Special attention should be paid to the following situations: operations other than normal standard operations (such as cleaning, waste disposal, repairs) should be carried out on the machine.

③ 机器所有者/操作者必须定期进行培训，使其了解产品生产中的潜在危险。

Machine owners/operators must receive regular training to be aware of potential hazards in product manufacturing.



### 3、机器的安全注意事项

#### Safety precautions for the machine

- a. 有关安全距离的信息，请参见 ISO13857.

For information on safety distances, please refer to ISO 13857.

- b. 有关防护装置其他信息，请参见如包装机械事故预防规定或其他有效的地方性规定。

For additional information on protective devices, please refer to regulations such as packaging machinery accident prevention regulations or other applicable local regulations.

### 4、工厂操作人员须知

#### Factory Operator Instructions

只有经过培训的熟练人员才能进行设置，保养和维修作业。

Only trained and skilled personnel are allowed to perform setup, maintenance, and repair operations.

操作动力装置的人员必须确保机器在运转时不会伤及自身和他人。

Personnel operating the power unit must ensure that the machine does not pose a risk to themselves and others while in operation.

动力装置的所有者/操作者可以指派具有以下能力的人员进行可能带来的危险的作业：有能力单独安全完成作业的人员，或在熟悉该作业人员的指导下并遵守以上所述操作说明进行作业的人员。

The owner/operator of the power unit may assign personnel with the

ability to perform potentially hazardous operations: personnel capable of safely completing the operation independently, or personnel who can perform the operation under the guidance of a knowledgeable operator and following the operating instructions mentioned above.

**4.1 如果由非熟练人员操作机器可能会发生危险。如果机器不按规定用途使用也会造成危险。**

**4.1 There may be dangers if the machine is operated by unskilled personnel. If the machine is not used for its intended purpose, it can also be dangerous.**

**1) 机器按照相关法律规定制造，在交货时确保为安全状态。**

The machine is manufactured in accordance with relevant laws and regulations, and is delivered in a safe condition.

任何残余危险在相关安全声明中有定义，并且在操作说明书里有描述。

Any residual risks are defined in the relevant safety statements and described in the operating instructions.

**2) 必须听从和遵守机器附加的安全信息。所有安全信息必须完整且容易理解。**

It is mandatory to comply with and adhere to the safety information provided with the machine. All safety information must be complete and easy to understand.

**3) 只有当所有的防护装置都准备就绪的情况下才允许操作机器。**

Machine operation is only allowed when all protective devices are

ready.

## 4.2 检查安全装置

### 4.2 Inspection of Safety Devices

4.2.1 安全装置，警报系统，锁定装置和包装连接器以及辅助包装设备每年必须至少由专业人员检测一次，以确保安全。

4.2.1 Safety devices, alarm systems, locking devices, packaging connectors, and auxiliary packaging equipment must be inspected at least once a year by professionals to ensure safety.

4.2.2 在第一次启动之前，专业人员应检测有害物质去除装置，确保其正常工作。

4.2.2 Before the initial start-up, professionals should inspect the harmful substance removal device to ensure its proper functioning.

4.2.3 以上所有检测结果必须记载在测试证明上，并保留到下次检测。

4.2.3 All the above inspection results must be recorded on a test certificate and kept until the next inspection.

## 4.3 指派负责人员

### 4.3 Assigning Responsible Personnel

4.3.1 只有技术熟练且经过培训的人员才能指派为负责人。而且必须明确其相关操作、装置、保养和维修作业的职责。

4.3.1 Only personnel who are technically proficient and trained can be assigned as responsible personnel. Their responsibilities for relevant operations, equipment, maintenance, and repair tasks must be clearly

defined.

4.3.2 定期检查相关人员是否遵守安全操作的规定，是否具备发现危险的意识。

4.3.2 Regularly check whether the relevant personnel comply with the safety operation regulations and have the awareness to identify hazards.

在开机前，相关人员应先阅读充分理解操作手册，尤其是有关“安全措施”的章节，并要求了解现行的有关规定。

Before starting up, the relevant personnel should read and fully understand the operation manual, especially the section on 'safety measures', and be required to understand the current relevant regulations.

操作手册和规定应放在操作人员和维护人员容易拿到的地方。

The operation manual and regulations should be placed in a location easily accessible to operators and maintenance personnel.

## 5、操作手册

### 5. Operation Manual

| 序号<br>Number | 措施<br>Measures                                                                                                                                                                                                  |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | 注意“电气/电子安装”中的提示信息<br>Pay attention to the tips in the 'Electrical/Electronic Installation' section.                                                                                                             |
| 2            | 按操作手册要求关闭机器，并确保不会意外启动。<br>Shut down the machine as required by the operation manual and ensure it cannot be accidentally started.                                                                               |
| 3            | 锁上主控制柜，拔掉钥匙/或在主开关上贴上警告标志<br>Lock the main control cabinet and remove the key/or attach a warning sign to the main switch.                                                                                       |
| 4            | 在有压缩空气，蒸汽压力和液压的装置上作业前，必须将压力发生器关上，并释放掉储存的气压。<br>Before working on devices with compressed air, steam pressure, and hydraulic pressure, the pressure generator must be shut off and the stored pressure released. |



|   |                                                                                                                                                |
|---|------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | 可移动部件比如工位的上部零件或可移动导轨必须有固定装置保护。<br>Movable components such as upper parts of workstations or movable guides must be protected by fixed devices. |
| 6 | 在作业前，必须等待所有发热部件都完全冷却。<br>Before starting work, all heated components must be completely cooled down.                                           |
| 7 | 工作区域必须特别注意安全措施（防护链，警告标志）<br>Special attention must be paid to safety measures in the work area (protective chains, warning signs).             |
| 8 | 通知工作人员，指派一人负责操作。<br>Notify the workers and assign one person to be in charge of the operation.                                                 |
| 9 | 取消错误。<br>Error Cancellation.                                                                                                                   |

## 6、电气/电子装置的安装及维护

### Installation and maintenance of electrical/electronic devices

注意：在电气部件/辅助设备上作业，必须由熟练电工遵照电气工程规定进行。如果接触到保险丝或其他通电的部件会危及生命。进行电气作业前，整条生产线必须同电源断开。可以关掉主开关/或拔掉主插头。

Note: When working on electrical components/auxiliary equipment, it must be carried out by skilled electricians following electrical engineering regulations. Contact with fuses or other energized parts can be life-threatening. Before performing electrical work, the entire production line must be disconnected from the power source. Turn off the main switch/unplug the main plug.

#### 6.1 重要操作规定

#### 6.1 Important Operating Regulations

- a. 定期检修机器，如果发现任和问题或错误必须立即解决。在问题未解决之前机器必须保持关机状态。

Regular maintenance of the machine is required, and any problems or errors found must be resolved immediately. The machine must remain powered off until the issue is resolved.

- b. 在电气安装/辅助设备/操作方式中发现的任何错误必须立即纠正。

Any errors found in electrical installation/auxiliary equipment/operation methods must be corrected immediately.

- c. 任何机器和部件在保养维护作业时，必须切断电源。

Power must be disconnected for any machine and parts during maintenance operations.

- d. 首先必须确认要维修部分的电源已经切断，并且确实没有电压，然后接地并使其短路，且必同周围有电流通过的部分绝缘。

First, ensure that the power to the section to be repaired has been disconnected and there is no voltage, then ground and short-circuit it, and insulate the parts with current passing through from the surroundings.

- e. 如果必须在导电的部件上作业，必须由另一人的帮助，在有紧急情况时，立即由其按下紧急停止开关或主开关来切断电源。

If it is necessary to work on conductive components, another person must assist. In case of emergency, the power must be immediately cut off by pressing the emergency stop switch or main switch.

- f. 作业时必须警戒作业区域，装上警告标志。只能使用绝缘电气工具。

Be vigilant in the work area and install warning signs during operation. Only insulated electrical tools should be used.

- g. 在高压辅助装置上作业时，断开电源后，应将电源线接地并将元件短路，比如用接地杆。

When working on high-voltage auxiliary devices, after disconnecting the power, the power cord should be grounded and the components short-circuited, for example, using a grounding rod.

- h. 保险丝不能修补或旁路。必须用制造商特定的保险丝。

Fuses cannot be repaired or bypassed. Manufacturer-specific fuses must be used.

## 6.2 严格遵守以下措施

### 6.2 Strictly adhere to the following measures:

1. 关掉机器主开关

Turn off the main power switch of the machine.

2. 将机器上的灰尘和可燃物质清除干净

Clean dust and combustible materials from the machine.

3. 不要将电焊设备插进或连接到机器控制柜上

Do not plug or connect welding equipment into the machine control cabinet.

4. 电子装置（例如控制装置等）和控制柜里的控制卡必须拔出，否

则会被瞬间高压损坏。

The control card in electronic devices (such as control devices, etc.) and control cabinets must be removed, otherwise it will be damaged by instant high voltage.

5. 机器上电子装备必须拔出或者终端断开

The electronic equipment on the machine must be unplugged or the terminal disconnected.

6. 电焊设备必须在电焊点旁接地，而且必须接地良好。

The welding equipment must be grounded near the welding point, and the grounding must be good.

## 7、安全装置及防护装置

## 7. Safety devices and protective devices

注意：只有在所有的防护装置都安装到位并能正常使用的情况下才能使用机器。

Note: The machine can only be used when all protective devices are installed properly and functioning correctly.

◆ 必要的消防措施，避免火灾发生。

Take necessary fire prevention measures to avoid fire hazards.

◆ 意外起火时，使用二氧化碳灭火器。

In case of accidental fire, use a carbon dioxide fire extinguisher.

本机器的制造符合现行的法律法规，可以安全使用。但是，如果操作



不当或在不适当的情况下操作，机器也会带来危险。

The manufacture of this machine complies with current laws and regulations and can be safely used. However, improper operation or operation under inappropriate conditions can also pose risks.

## 7.1 防护装置

### 7.1 Protective devices

1. 机器上危险的地方标有警告标志。

Warning signs are placed in dangerous areas on the machine.

2. 可视的错误指示。

Visible error indication.

3. 工作流程的自动控制在出现错误后自动中断运行。

Automatic control of the workflow will be automatically interrupted in case of errors.

4. 紧急停止按钮。

Emergency stop button.

5. 如果工作区域的保护装置被打开，机器会自动停止。

If the protective device in the work area is opened, the machine will stop automatically.

6. 控制柜机械连动主开关。严禁用水枪或高压清洗设备冲洗控制柜。

Control cabinet mechanical interlock main switch. It is strictly prohibited to wash the control cabinet with a water gun or high-pressure cleaning equipment.

7. 排气装置和风扇。

Exhaust device and fan.

8. 机器上需要观察的区域装有透明观察窗。

Areas on the machine that need to be observed are equipped with transparent observation windows.

## 8、加工粉尘的产品

### 8. Products processing dust.

未过滤物质极有可能引起中毒危险

Substances that are not filtered out may pose a poisoning risk.

过滤器不能消除所有的危险颗粒。处理有毒产品可能导致严重中毒。

Filters cannot eliminate all hazardous particles. Handling toxic products may result in severe poisoning.

◆ 所有过滤器的使用均应参照其说明书规定（效率/微粒大小）

All filter usage should comply with the instructions provided (efficiency/particle size).

◆ 运营公司有责任检查过滤器是否适合自身所加工的产品。

Operating companies are responsible for checking if the filters are suitable for the products they process.

粉尘可能导致爆炸危险。爆炸可由以下状况触发：

Dust may cause explosion hazards. Explosions can be triggered by the following conditions:

◆ 火花，例如拉拔电气插头。

Sparks, such as pulling electrical plugs.

◆ 电气设备的错误应用。

Improper application of electrical equipment.

释放静电的衣服。要及时移除累积的粉尘，定期检查电气设备。

Clothing that releases static electricity. To remove accumulated dust in a timely manner and inspect electrical equipment regularly.

## 三、机器相关详细内容

### III. Detailed content related to the machine

#### 1.1 主要技术参数及辅机的选购

#### 1.1 Main technical parameters and selection of auxiliary machines

主要技术参数 Main Technical Parameters

|                           |                           |
|---------------------------|---------------------------|
| 产 量 Output                | 3500 粒/分 ( Capsules/min ) |
| 模孔数量 No.of segment bores  | 25                        |
| 运用胶囊 Suitable for capsule | 000#-5#                   |
| 总 功 率 Total power         | 10.5Kw                    |
| 主机重量 Overall weight       | 2200Kgs                   |
| 主机尺寸 Overall dimensions   | 1555mm*1165mm*2040mm      |

#### 该机器有以下标记的可选设备

The machine has the following optional equipment marked:

自动上粉装置

Automatic powder feeding device

微丸灌装装置

Microgranule filling device

加强清洁模块

Enhanced cleaning module

胶囊上料机

Capsule loading machine

抛光机

Polishing machine

## 1.2 工作原理及其主要结构

## 1.2 Working Principle and Main Structure

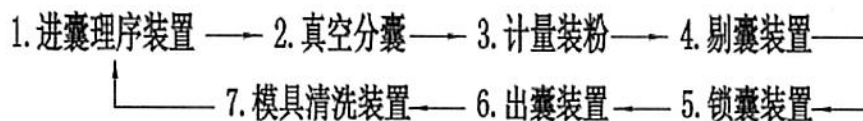
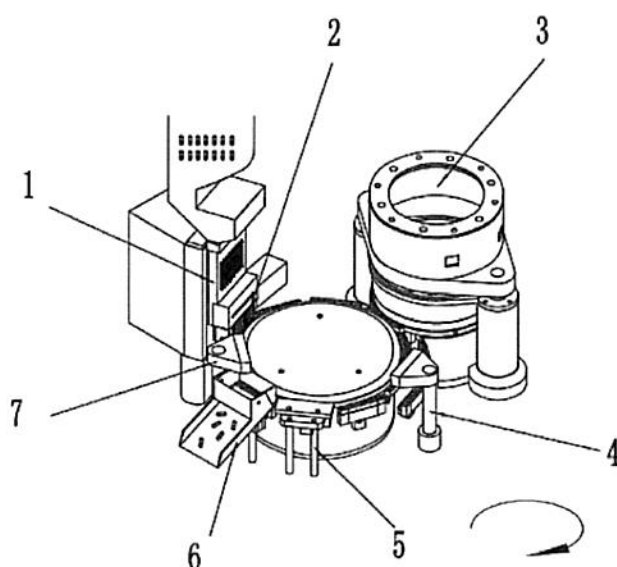
### 1.2.1 工作原理

#### 1.2.1 Working Principle

本机胶囊填充工作过程如下：

The working process of the capsule filling machine is as follows:

工位运转流程



工位运转流程

Station operation process

1.进囊理序装置

1. Capsule sorting device

2.真空分囊

2. Vacuum dividing

3. 计量装粉

3. Metering and powder filling

4. 剔囊装置

4. Pouching device

7. 模具清洗装置

7. Mold cleaning device

6. 出囊装置

6. Outpouching device

5. 锁囊装置

5. Locking device

### 1.2.2 上体执行工作部分

#### 1.2.2 Upper body working part

由播囊理序组件、工位转盘、充填组件、胶囊斗、粉料斗、剔囊组件、锁囊组件、出囊组件、真空分囊组件等结构组成。

It is composed of components such as capsule sorting component, workstation turntable, filling component, capsule hopper, powder hopper, capsule picking component, capsule locking component, capsule discharging component, vacuum capsule separating component.

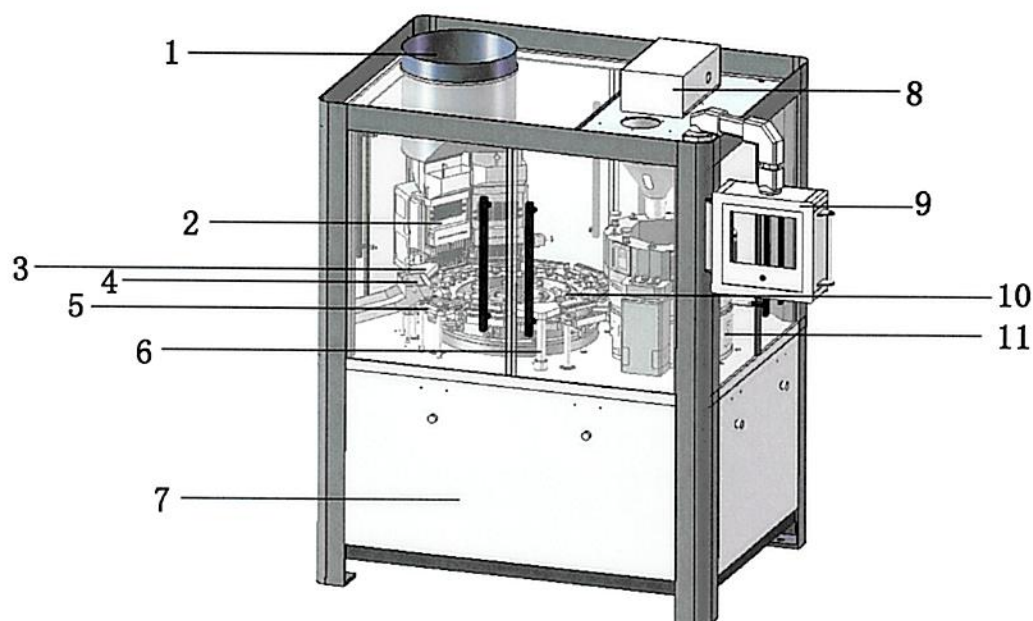
### 1.2.3 下体是机械传动部分及动力部分

#### 1.2.3 The lower body is the mechanical transmission part and power part



由传动机构、传动链、动力源等机构组成。

It is composed of mechanisms such as transmission mechanism, transmission chain, power source, etc.



1.胶囊斗 2. 播囊装置 3.清洗装置 4.出料装置 5.锁囊装置 6.剔废装置 7.传动部分 8.加料电机箱及粉料斗 9.触摸屏 10.工位转盘

1. Capsule hopper 2. Capsule loading device 3. Cleaning device 4. Discharging device 5. Capsule locking device 6. Rejection device 7. Transmission part 8. Feeding motor box and powder hopper 9. Touch screen 10. Workstation turntable

11. 充填装置

11. Filling device



#### 1.2.4 气动部分

#### 1.2.4 Pneumatic part

本机利用真空泵吸气进行胶囊帽体分离、用工业吸尘器进行废囊粉尘清理工作。

The machine uses a vacuum pump for capsule cap separation and an industrial vacuum cleaner for waste capsule dust cleaning.

#### 1.2.5 控制部分

#### 1.2.5 Control part

本机电气装置采用液晶式显示屏和电脑（PLC）控制技术，可进行机器运行状态的显示和预置，并显示故障现象、原因判断和排除方法等“人机对话”功能，采用美国艾默生变频控制技术，对整机进行无级调速。

The local electrical device adopts a liquid crystal display screen and computer (PLC) control technology, which can display and preset the machine's operating status, display fault phenomena, reasons for judgment, and troubleshooting methods, etc. It has a "human-machine dialogue" function. It adopts Emerson's frequency conversion control technology from the United States to achieve stepless speed regulation of the whole machine.





主机点动

Jogging of the host

料斗升

Hopper Up

开胶囊

Open the capsule

主机风机点动

Host fan jog

料斗降

Hopper descent

关胶囊

Capsule close

上胶囊

Capsule on

停止

Stop

## 1.3 安装与试车

## 1.3 Installation and Commissioning

### 1.3.1 安装

### 1.3.1 Installation

### 1.3.2 主机定位

### 1.3.2 Main Machine Positioning

本机搬运、安置应轻放，避免机内电脑元件等损失。机器定位时，  
The machine should be handled and placed lightly during transportation  
and installation to avoid damage to internal computer components. When  
the machine is positioned,

底座下方必须用橡皮板垫实，用以避震。台面要求水平且不得有摆动  
现象。

Rubber pads must be placed under the base to avoid vibration. The  
countertop must be level and free of any swinging phenomenon.

### 1.3.3 环境要求

### 1.3.3 Environmental Requirements

环境温度：21±3℃

Environmental temperature: 21±3℃

相对湿度：45—50%

Relative humidity: 45-50%

海拔高度：1000 米以下

Altitude: below 1000 meters

#### 1.3.4 辅机安装建议

#### 1.3.4 Installation Recommendations for Auxiliary Equipment

本机辅机—真空泵、吸尘器建议与主机隔离安装，减少噪声，避免长期噪音对操作者产生不适及身心危害。

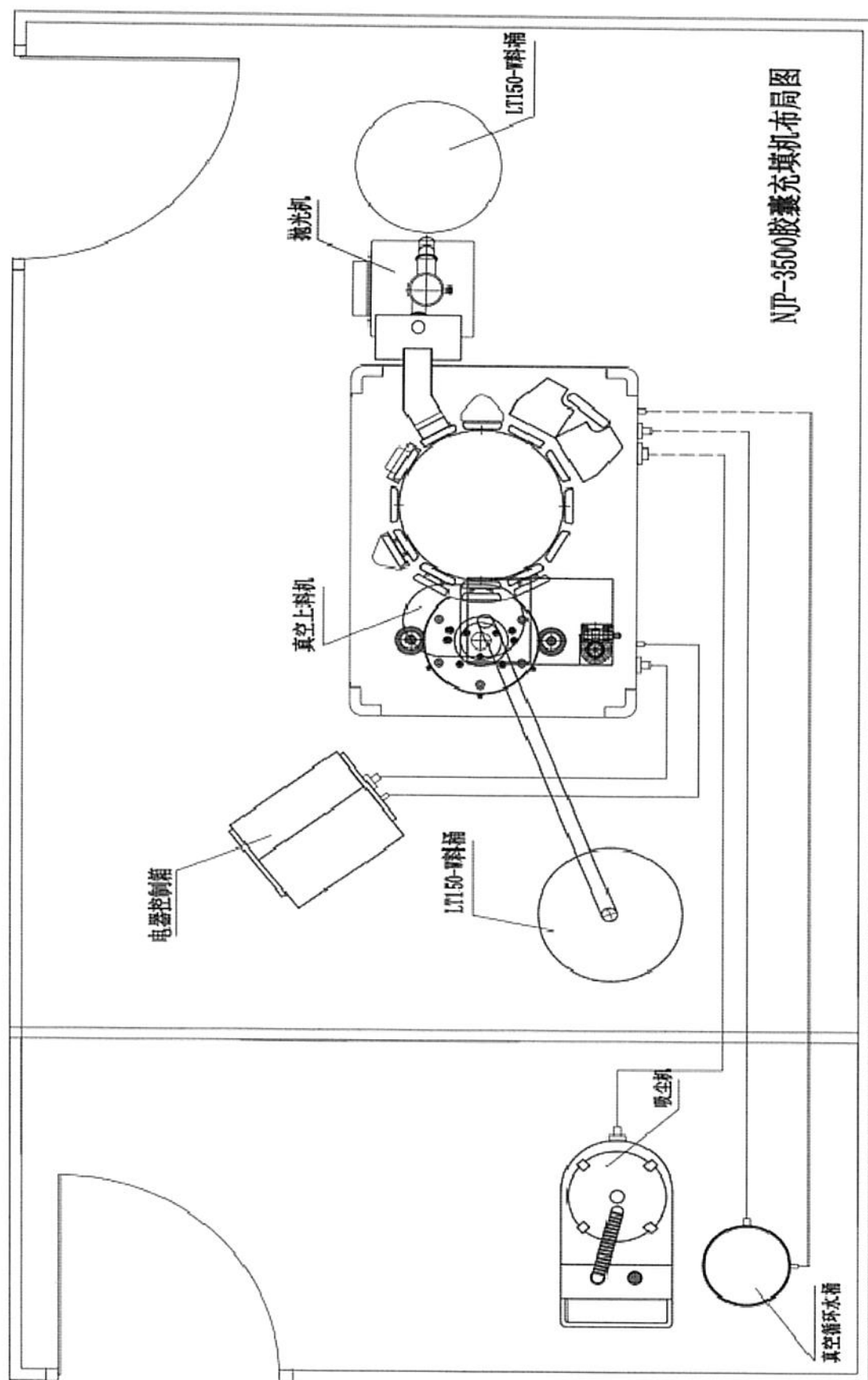
It is recommended to install the auxiliary equipment such as vacuum pumps and vacuum cleaners separately from the main machine to reduce noise, avoid long-term discomfort and physical and mental harm to operators caused by noise.

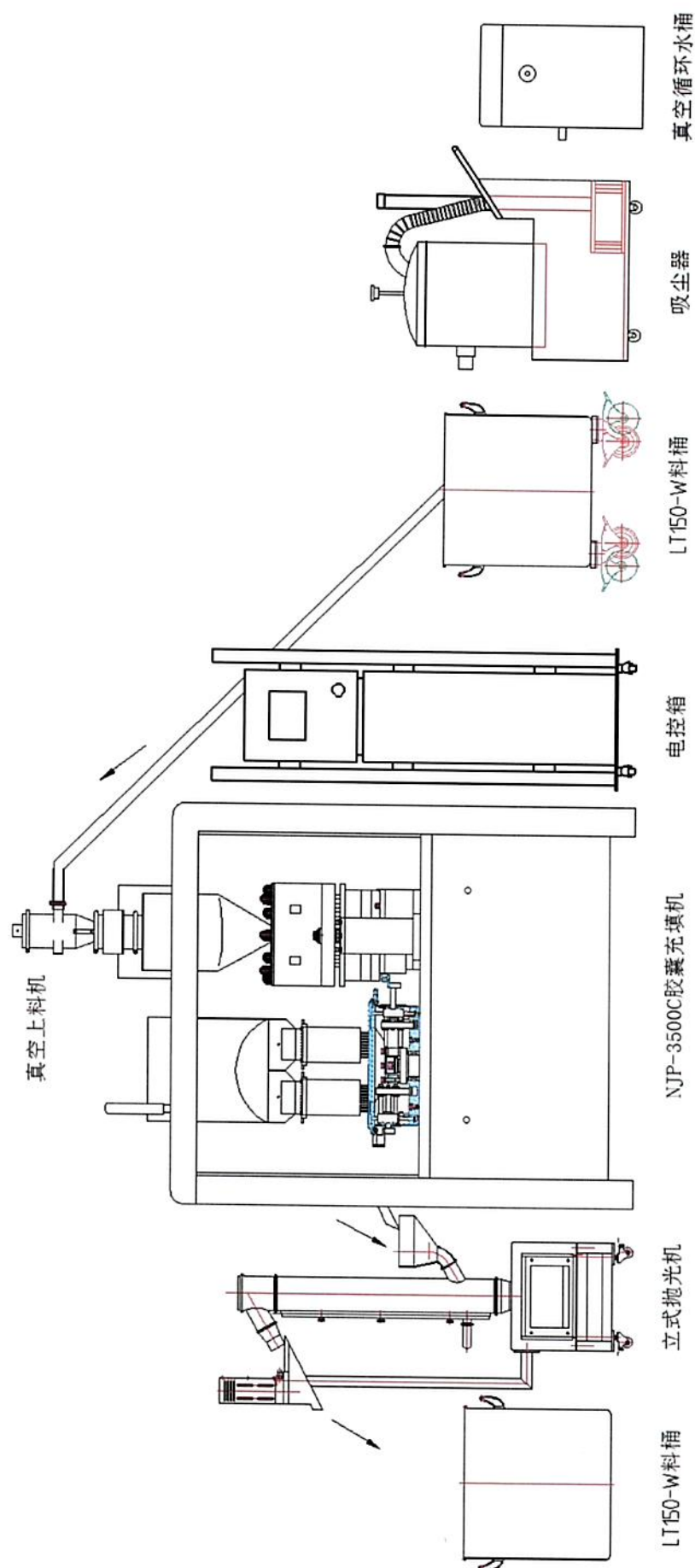
#### 1.3.5 连接真空泵、吸尘器主体气管。

#### 1.3.5 Connect the main air pipe of the vacuum pump and vacuum cleaner.

#### 1.3.6 接上电源，应注意电压与主机要求符合。

#### 1.3.6 Connect the power supply, pay attention to the voltage requirements matching the main machine.





NJP-3500C胶囊充填机立面流程图



电源控制箱

Power control box

电控箱

Electric control box

抛光机

Polishing machine

立式抛光机

Vertical polishing machine

料筒

Material barrel

胶囊充填机

Capsule filling machine

胶囊充填机布局图

Layout diagram of capsule filling machine

吸尘器

Vacuum cleaner

真空循环水箱

Vacuum Circulating Water Tank

## 2 试车

### 2 test drive

2.1 检查整机各部位螺栓、螺钉、螺母、销钉等紧固件，有无松动脱落现象（在运输、搬运、安装过程中出现的振动引起）。

2.1 Check all parts of the machine for loose or missing bolts, screws, nuts, pins, etc. (caused by vibration during transportation, handling, and installation).

2.2 将手轮套入主电机尾部轴颈上，手动盘车，使工位部分运转 3-5 周，并观察各零部件是否协调正常。

2.2 Insert the handwheel into the rear shaft neck of the main motor, manually rotate the handwheel, let the workstation run for 3-5 cycles, and observe if all components are operating normally.

2.3 接通电源开关，参照机器的启动与操作规程进行开机空运转 1—2 小时。

2.3 Turn on the power switch, follow the machine's startup and operation procedures to run the machine empty for 1-2 hours.

## 1.4 机器各部件的微调整及其说明

### 1.4 Fine adjustment of machine components and their instructions

#### 1.4.1 空胶囊料斗闸门的调整

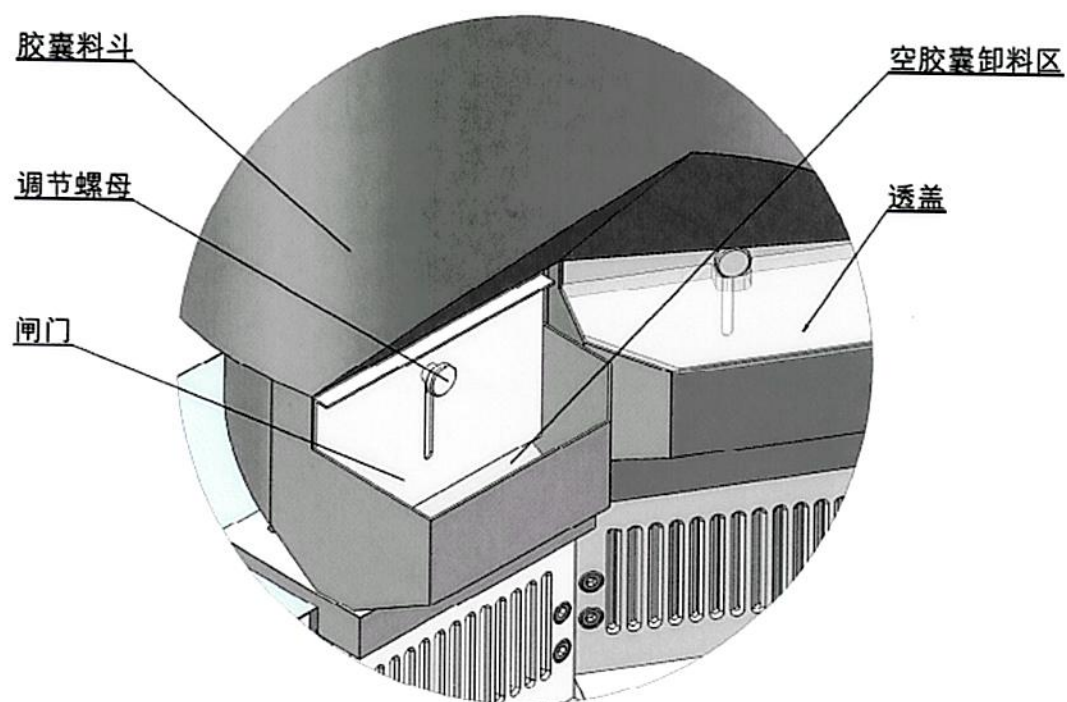
##### Adjustment of Empty Capsule Hopper Gate

胶囊料斗组件由胶囊料斗、闸门、玻璃透盖、调节螺母等零部件组成。闸门高度由调节螺母固紧控制，调节高度可以改变空胶囊卸料区的卸料值；空胶囊的堆积高度，约为总出口高度的  $1/2$ 。

The capsule hopper assembly consists of components such as the capsule hopper, gate, glass cover, and adjusting nut. The gate height is controlled by tightening the adjusting nut, and adjusting the height can change the unloading value of the empty capsule unloading area; the stacking height of empty capsules is about  $1/2$  of the total outlet height.

(注意：生产工作后，若空胶囊卸料区尚有剩余空胶囊，应当及时清理。避免下次工作时出现生产异常)

(Note: After production work, if there are remaining empty capsules in the unloading area, they should be cleaned up promptly. This is to avoid production abnormalities in the next work.)



胶囊料斗

Capsule hopper

空胶囊卸料区

Empty capsule unloading area

调节螺母

Adjusting nut

透盖

Permeable cover

闸门

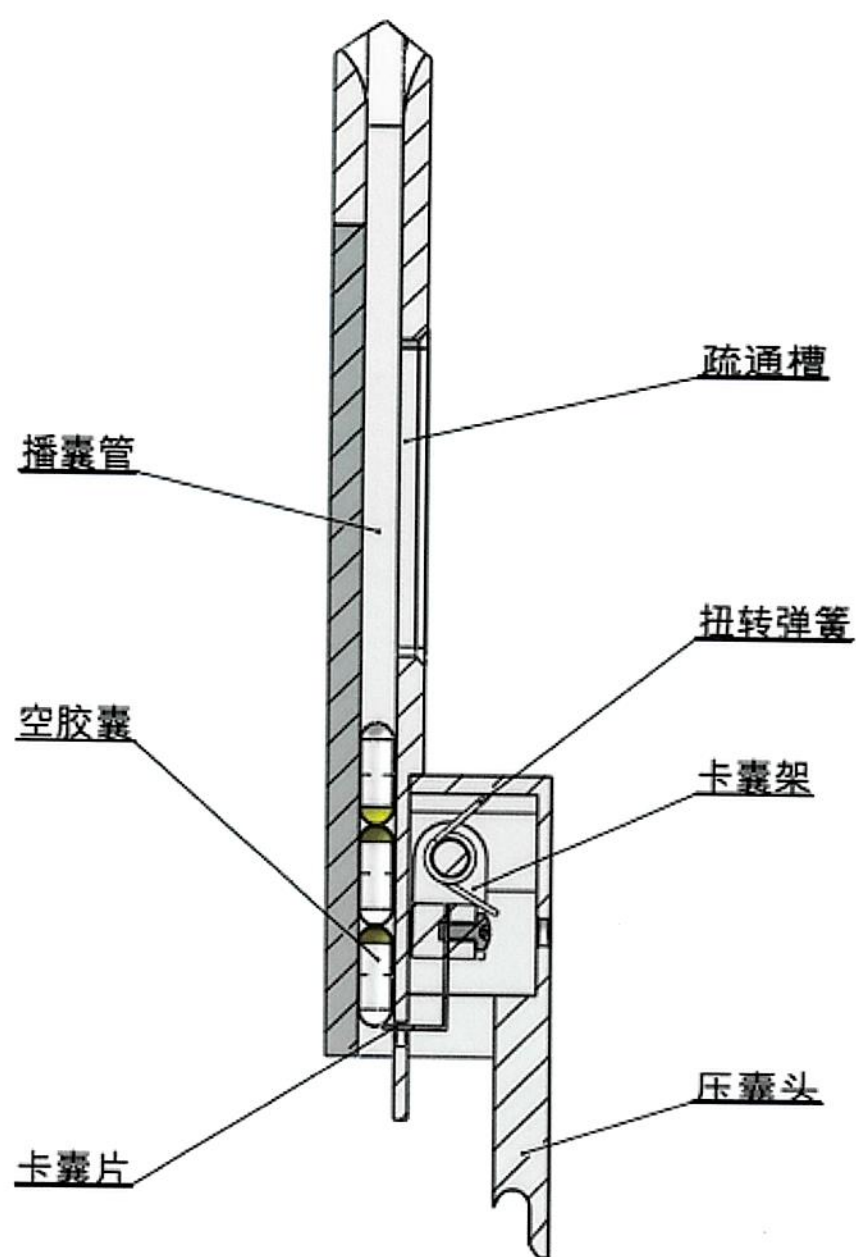
Gate

#### 1.4.2 卡囊片的调整

#### 1.4.2 Adjustment of capsule strip

卡囊片压入深度由手工调节、确保每次播囊一颗、后颗胶囊被卡囊片锁在管道内。

The depth of capsule strip insertion is manually adjusted to ensure that one capsule is dispensed each time, and the subsequent capsules are locked in the tube by the capsule strip.



播囊管

Feeding tube

空胶囊

Empty capsule

卡囊片

Carded tablet

疏通槽

Clearing groove

扭转弹簧

Twist spring

卡囊架

Cartridge rack

压囊头

Capsule head

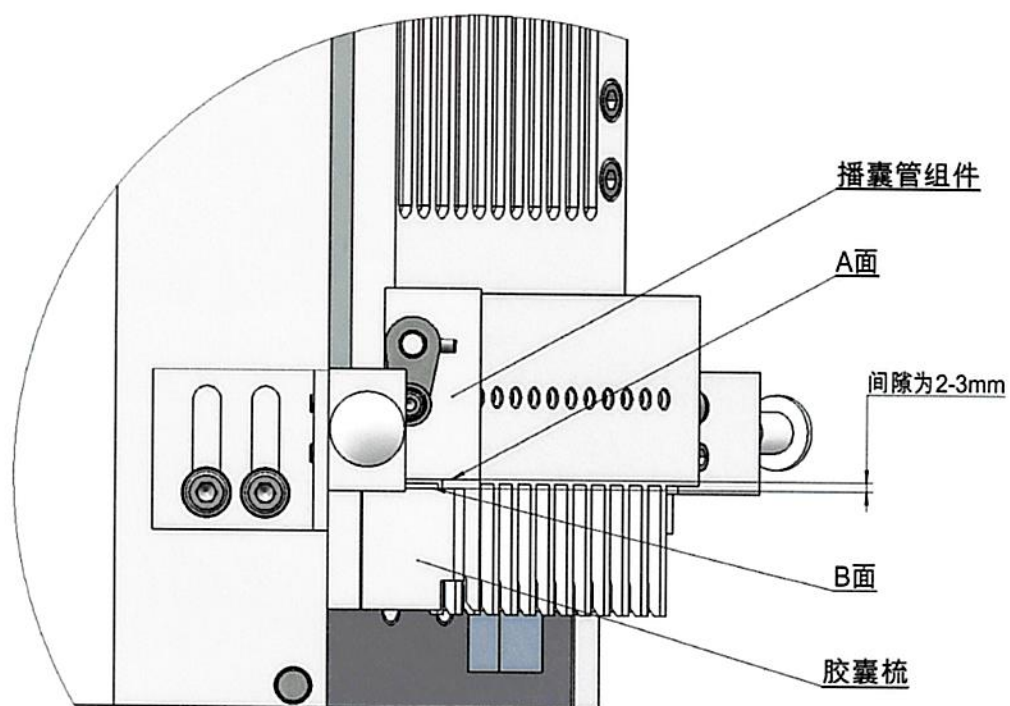


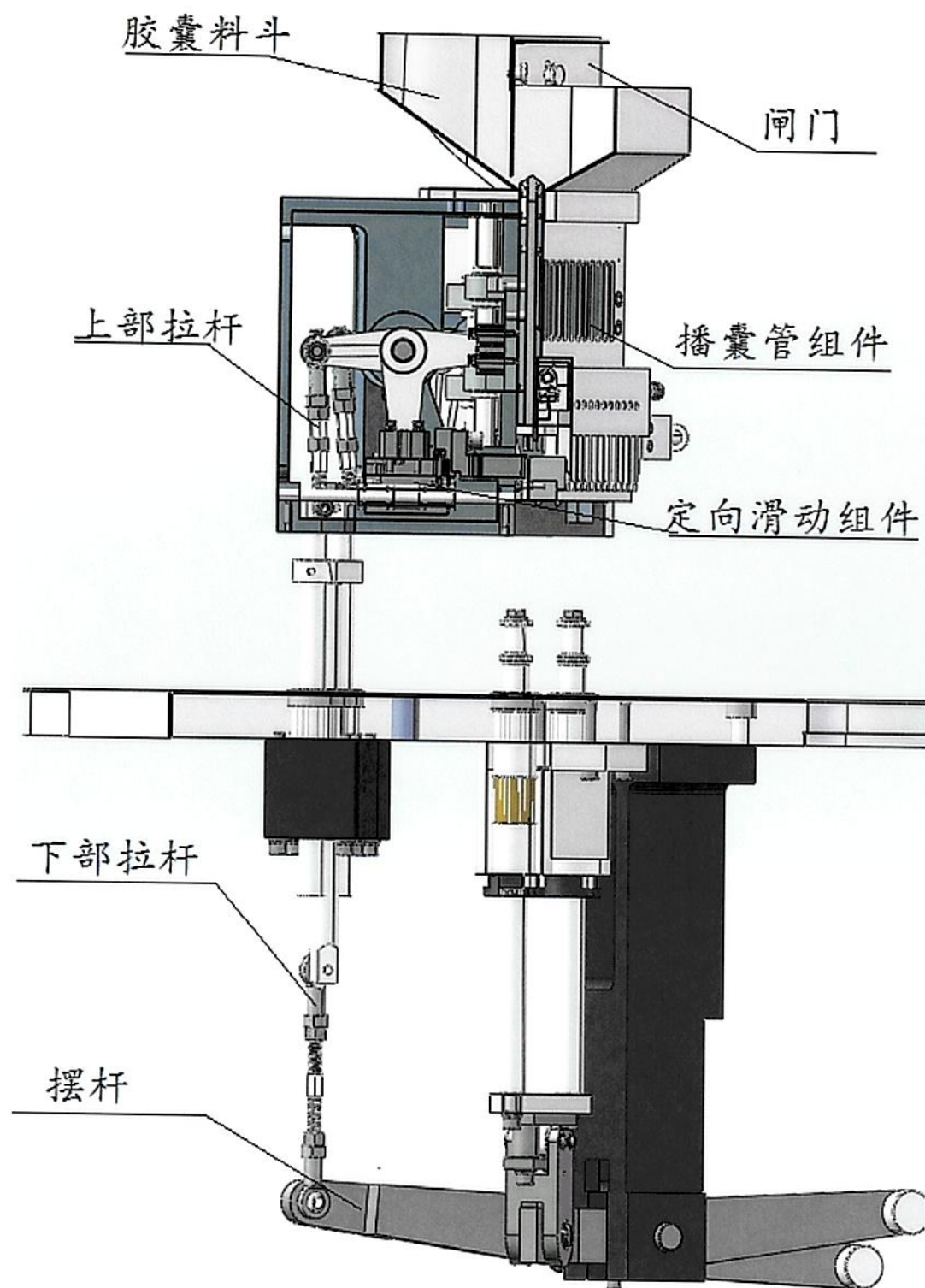
### 1.4.3 播囊机构的调整

#### 1.4.3 Adjustment of capsule dispensing mechanism

播囊机构实行进囊、贮囊、倒序、出囊、推囊等往复运动，播囊管组件 A 面与胶囊梳 B 面的间隙为 2—3mm。间隙差由机体下部拉杆来控制，利用附件工具中的对应扳手来调节拉杆（二头螺）长度

The capsule dispensing mechanism performs reciprocating movements such as capsule insertion, storage, reverse order, dispensing, and pushing. The gap between the A surface of the capsule dispensing tube assembly and the B surface of the capsule comb is 2-3mm. The gap difference is controlled by the lower rod of the body, and the length of the rod (double-headed screw) is adjusted using the corresponding wrench in the accessory tools.





播囊管组件

Capsule tube assembly

A 面

A FACE

间隙为 2-3mm

Gap is 2-3mm

B 面

B FACE

胶囊梳

Capsule comb

胶囊料斗

Capsule hopper

闸门

Gate

上部拉杆

Upper pull rod

播囊管组件

Capsule tube assembly

F 定向滑动组件

F directional sliding component

下部拉杆

Lower pull rod

摆杆

Swing arm

1.4.3.1 胶囊倒序

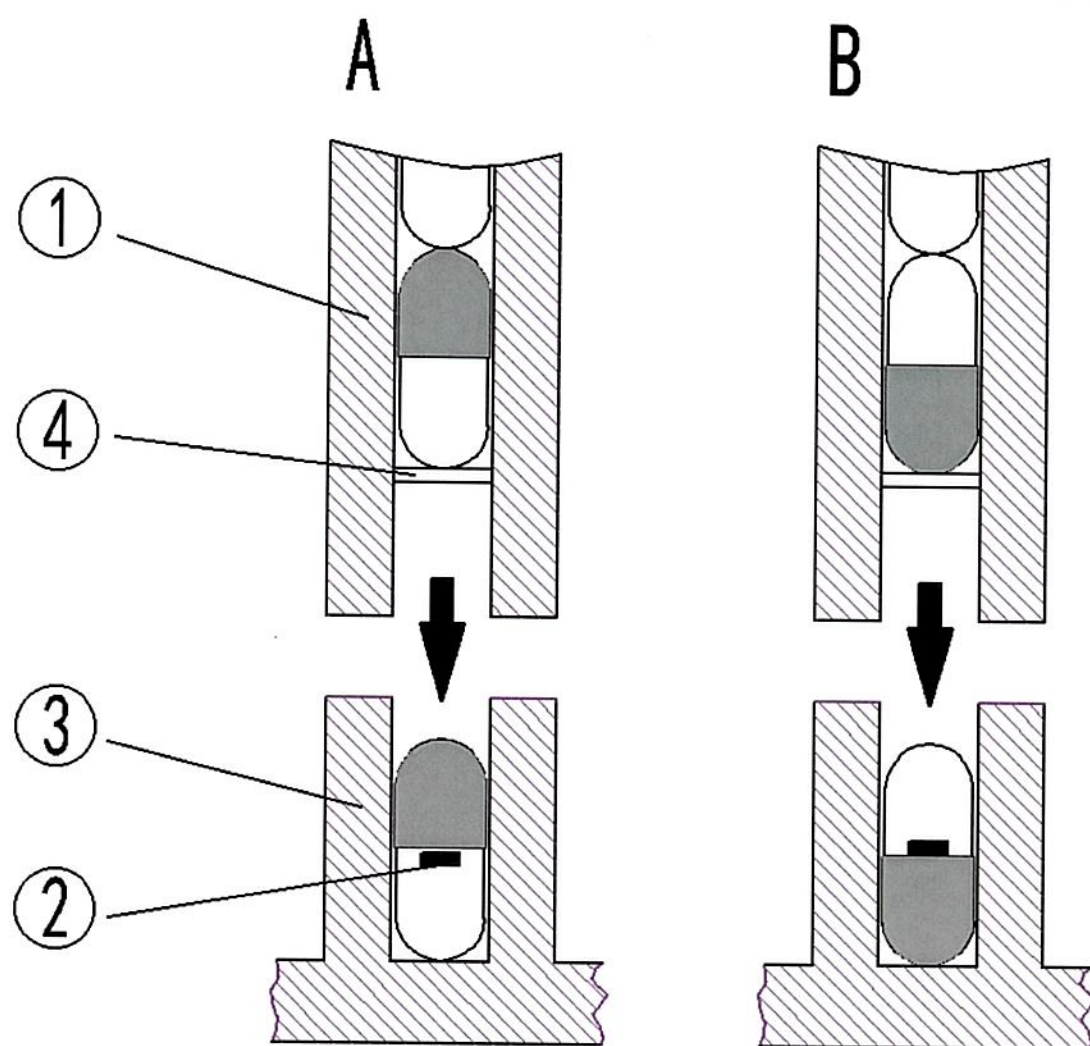
1.4.3.1 Capsule reverse order

胶囊在胶囊槽 1 中将出现如下图 A 胶囊帽向上和 B 胶囊帽向下两种情况。

The capsule will appear in two situations in capsule groove 1: A capsule cap facing up and B capsule cap facing down.

当胶囊槽 1 向下运动时，卡囊片 4 转动，下放一颗胶囊进入排序块 3 中。A 情况时排序推杆 2 与胶囊接触点在胶囊帽下方，B 情况则在上方。

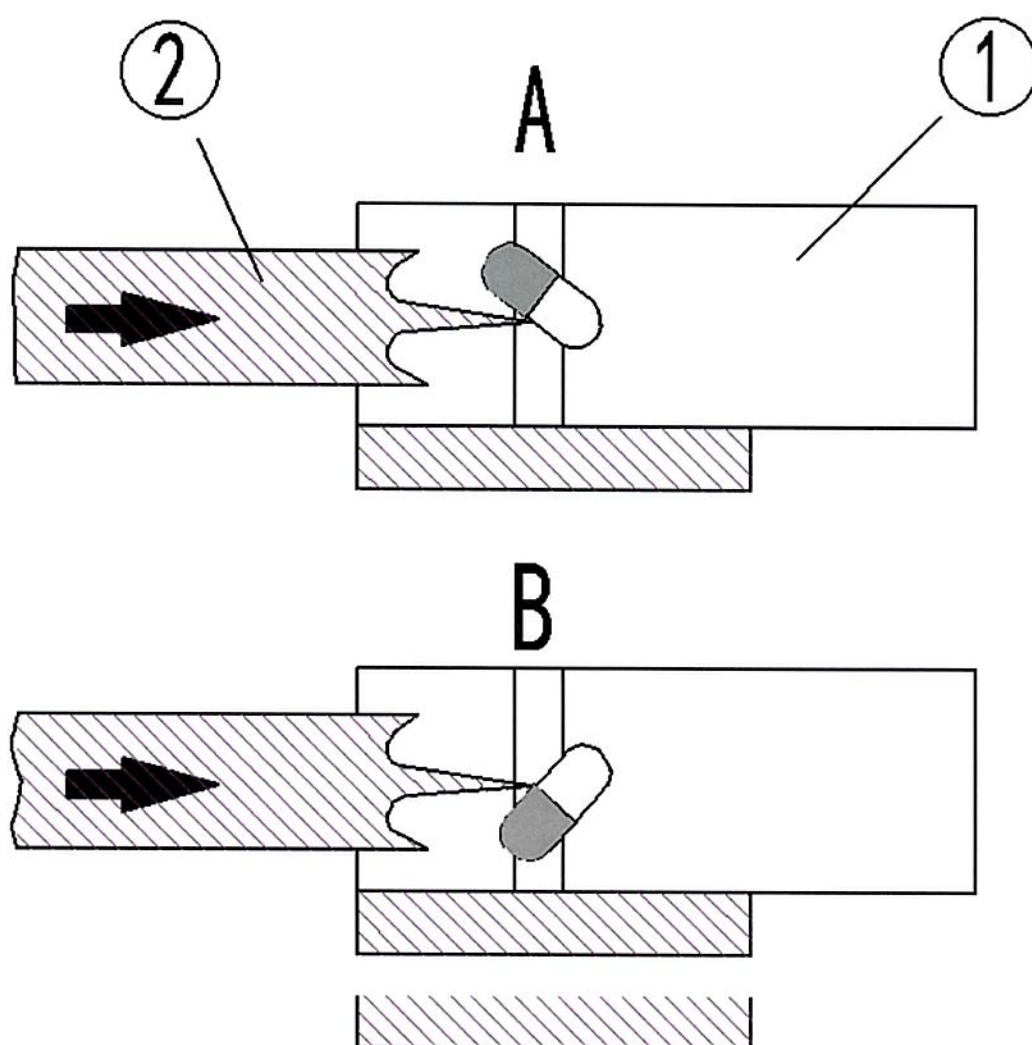
When capsule groove 1 moves downward, the capsule piece 4 rotates and drops a capsule into the sorting block 3. In situation A, the sorting push rod 2 contacts the capsule below the capsule cap, while in situation B, it contacts above.





当排序推杆 2 推动胶囊时将出现如下图两种状态：

When the sorting push rod 2 pushes the capsule, two states as shown in the figure below will appear:



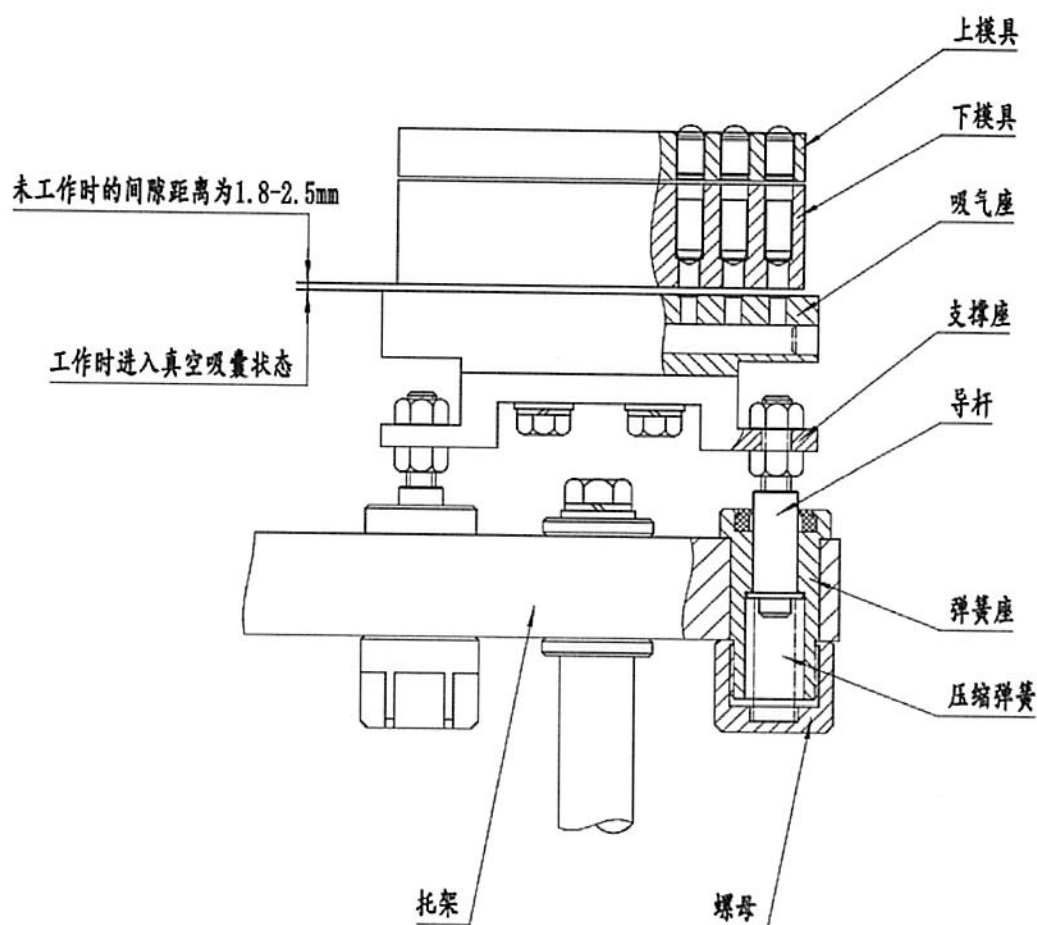


#### 1.4.4 真空分囊机构

#### 1.4.4 Vacuum Segmentation Mechanism

真空分离机构位于播囊工位下方，它由吸气座、支撑座、弹簧座、压缩弹簧、托架等零部件组成。在未工作时，吸气座上表面与模具下表面间隙为 1.8—2.5mm,工作时，吸气座上表面与模具下表面完全贴合进入真空吸囊状态。利用附件工具中的对应扳手来调整下部调节拉杆可解决吸气座与模具间隙差的问题。

The vacuum separation mechanism is located below the capsule station and consists of components such as the suction seat, support seat, spring seat, compression spring, and bracket. When not in operation, the gap between the upper surface of the suction seat and the lower surface of the mold is 1.8-2.5mm. During operation, the upper surface of the suction seat completely fits into the vacuum suction state with the lower surface of the mold. Adjusting the lower adjustment rod with the corresponding wrench in the attachment tool can solve the problem of the gap between the suction seat and the mold.



未工作时的间隙距离为 1.8-2.5mm

The gap distance when not working is 1.8-2.5mm.

工作时进入真空吸囊状态

Entering the vacuum suction state during operation.

上模具

Upper mold

下模具

Lower mold

吸气座

Suction seat

支撑座

Support seat

导杆

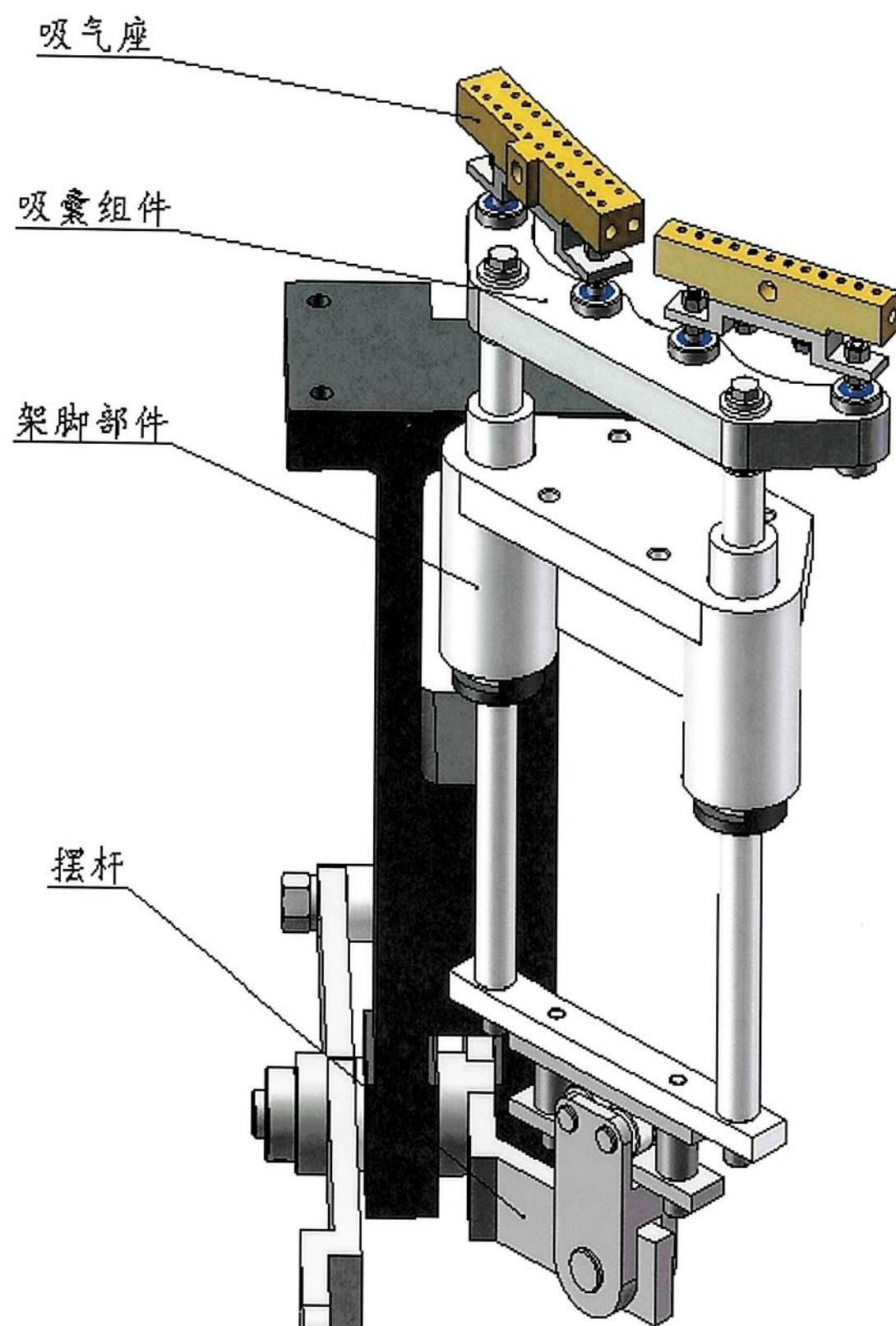
Guide pin

弹簧座

Spring seat

压缩弹簧

Compression spring



吸气座

Suction seat

吸囊组件

Suction bag assembly

架脚部件

Bracket foot component

摆杆

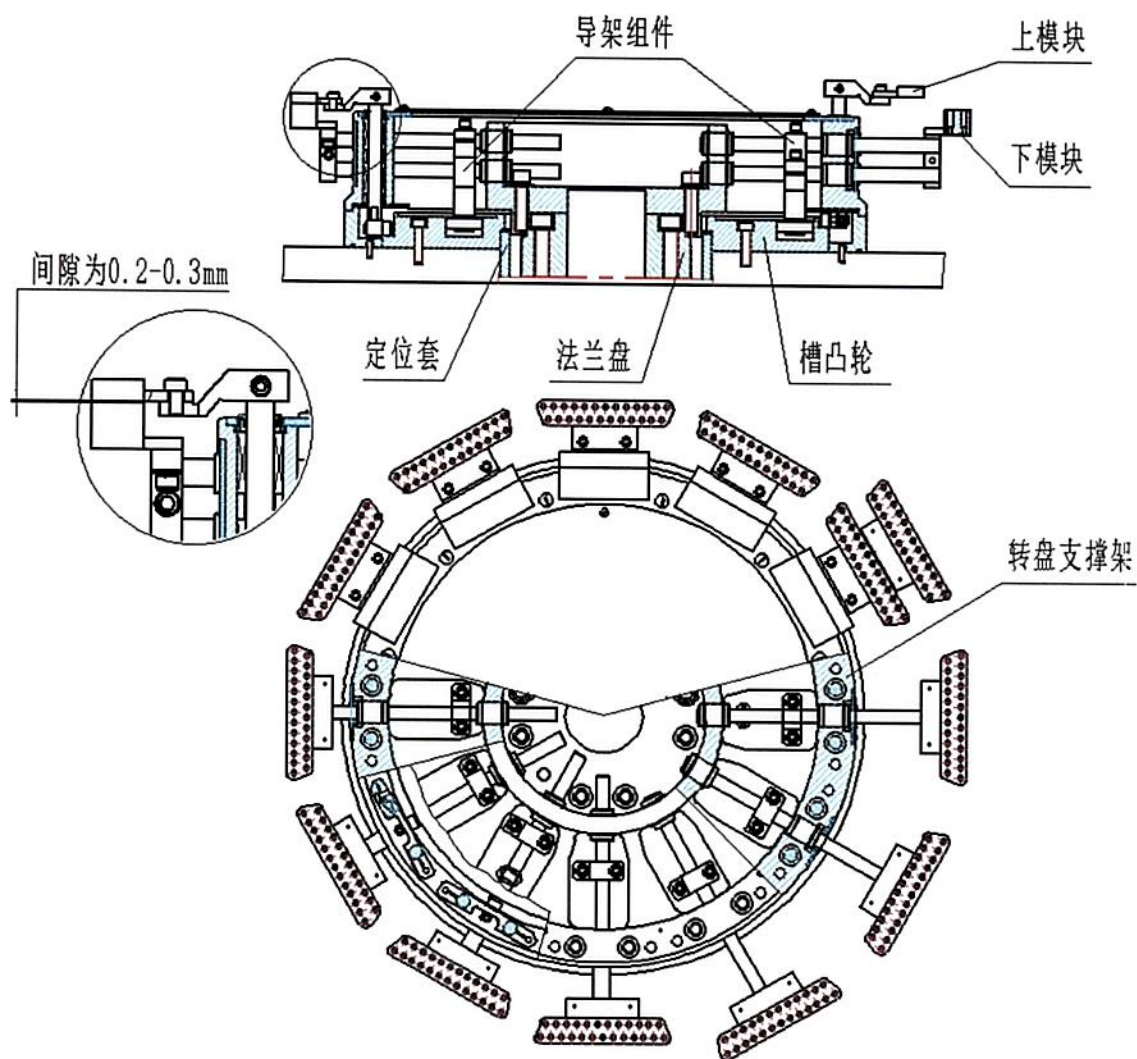
Swing arm

#### 1.4.5 工位转盘—模具的校准及清洗

Calibration and cleaning of the 1.4.5 station turntable-mold.

a. 工位转盘机构由上下模块、导架组件、转盘支撑架、槽凸轮、法兰盘、定位套等主要零部件组成。其上模块作圆周上下运动，下模块作圆周进退运动，上下模之间的间隙利用附件工具中的塞尺片校准控制为 0.2—0.3mm。

a. The station turntable mechanism is mainly composed of upper and lower modules, guide frame components, turntable support frame, groove cam, flange plate, positioning sleeve, etc. The upper module moves in a circular motion up and down, while the lower module moves in a circular motion back and forth. The gap between the upper and lower modules is calibrated using the feeler gauge in the accessory tools to be 0.2-0.3mm.



导架组件

Guide frame assembly

上模块

Upper module

下模块

Lower module

间隙为 0.2-0.3mm

Gap is 0.2-0.3mm



定位套,

Positioning sleeve

法兰盘

Flange plate

槽凸轮

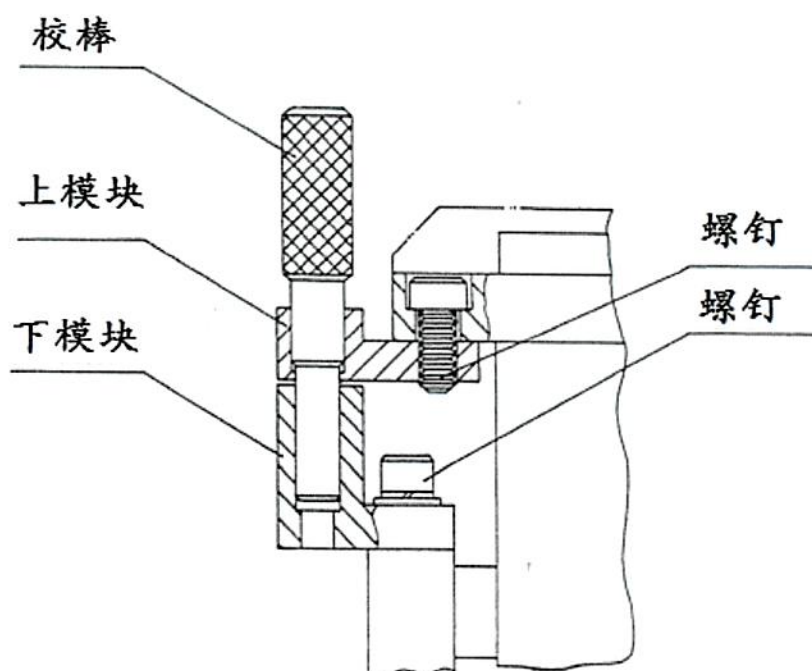
Camshaft

转盘支撑架

Turntable support bracket

b.校准要求，采用附件工具中的二条校棒校准模孔同心度，要求在0.01—0.02mm。(具体方法参见 1.6.1 )

b. Calibration requirements: Use the two calibration rods in the accessory tools to calibrate the concentricity of the die holes, with a requirement of 0.01-0.02mm. (Refer to section 1.6.1 for specific methods)



校棒  
Alignment pin  
上模块  
Upper module  
螺钉螺钉  
Screw bolt  
下模块  
Lower module

c. 模具清洗，不必拆卸转盘，因工位转盘为全密封组合设计，该机构是同类产品中改进后的最佳部分。

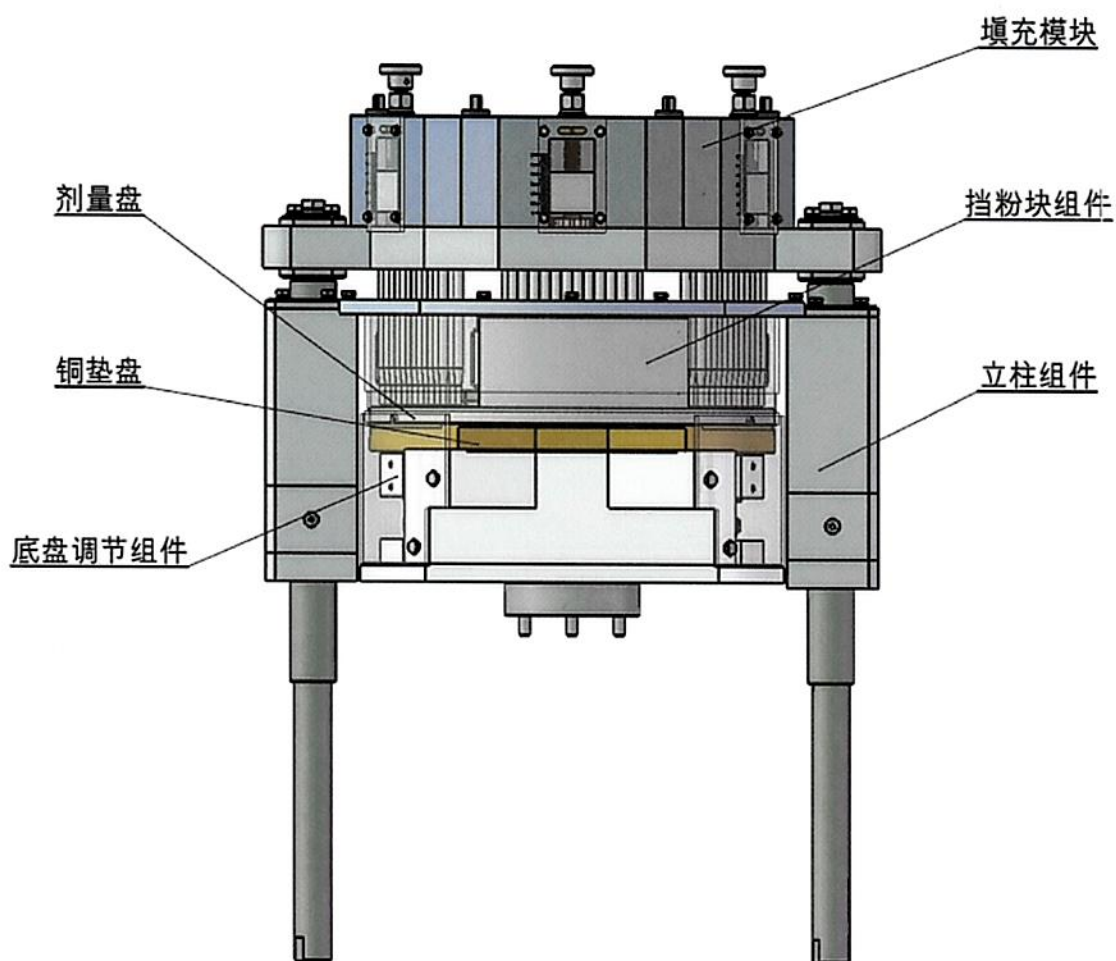
c. Mold cleaning: There is no need to disassemble the turntable, as the work station turntable is designed as a fully sealed assembly, which is the best improved part among similar products.

#### 1.4.6 剂量盘与密封环间隙调整

#### 1.4.6 Adjustment of the gap between the dosing disc and the sealing ring

本机粉料填充部分由填充模块、挡粉组件、剂量盘、铜垫盘、底盘调节组件及立柱组件等组成。

The native powder filling section is composed of filling module, powder blocking component, dosing disc, copper pad, base plate adjustment component, and column component, etc.



填充模块

Filling module

剂量盘

Dose plate

挡粉块组件

Powder blocking component

铜垫盘

Copper gasket

立柱组件

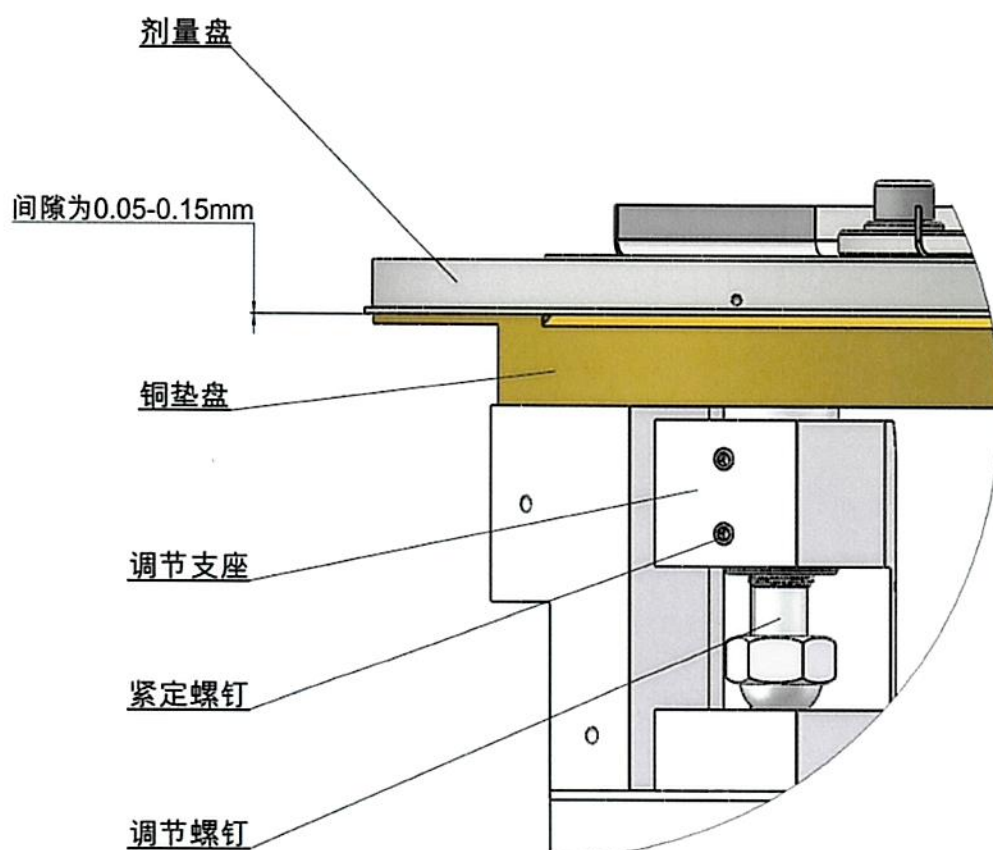
Column assembly

底盘调节组件

Base plate adjustment component

剂量盘与铜垫盘之间的间隙利用附件工具中的塞尺片校准控制为  
0.05—0.15mm。

The gap between the dosing disc and the copper pad is calibrated to  
0.05-0.15mm using the feeler gauge in the accessory tool.



剂量盘

Dose plate

间隙为 0.05-0.15mm

Clearance is 0.05-0.15mm

铜垫盘

Copper gasket

调节支座

Adjustable support

紧定螺钉

Locking screw

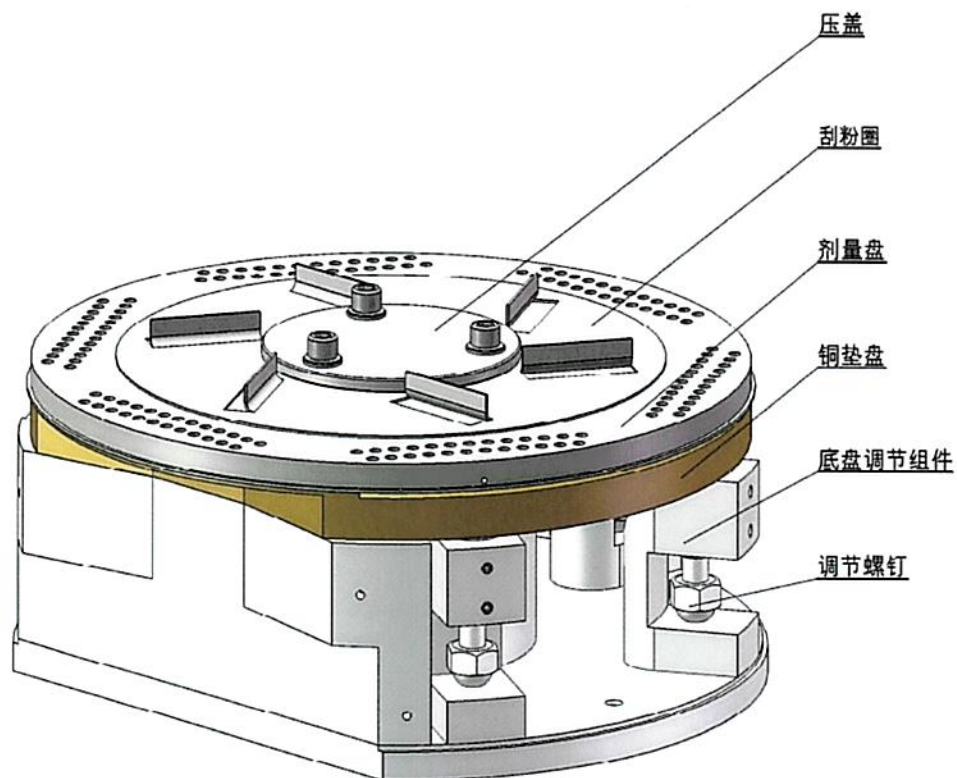
调节螺钉

Adjusting screw



调整方法：拆下填充模块（详见 1.6.2.1）、挡粉块组件、及密封圈。松开调节支座上的紧定螺钉，转动调节螺钉，利用塞尺片均匀调节铜垫盘水平面。完成后，固紧紧定螺钉，再次用塞尺片重复检测一次（该机构特点：以剂量盘下平面为基准，三维调节机构，间隙均匀，计量差异稳定，药粉利用率高，拆洗安装方便，为国内独创）。达到要求后，打开传动部分的侧门板，取出附件工具中的手轮，套入主电机输入轴，（顺时针）手动转动电机输入轴，以手感转动均匀为准。

Adjustment method: Remove the filling module (see 1.6.2.1), powder blocking component, and sealing ring. Loosen the fixing screw on the adjustment bracket, rotate the adjustment screw, and adjust the level of the copper pad evenly using the feeler gauge. After completion, tighten the screws and repeat the measurement with a feeler gauge (Features of this mechanism: based on the flat surface under the dose tray, three-dimensional adjustment mechanism, uniform clearance, stable dose difference, high powder utilization rate, easy disassembly, cleaning, and installation, unique domestically). Once the requirements are met, open the side door of the transmission part, take out the handwheel from the accessory tools, fit it into the main motor input shaft, and manually rotate the motor input shaft (clockwise) until it rotates smoothly by feel.



压盖

Pressure cover

刮粉圈

Powder scraper

剂量盘

Dose plate

铜垫盘

Copper gasket

底盘调节组件

Base plate adjustment component

调节螺钉

Adjusting screw

#### 1.4.7 挡粉器间隙调整

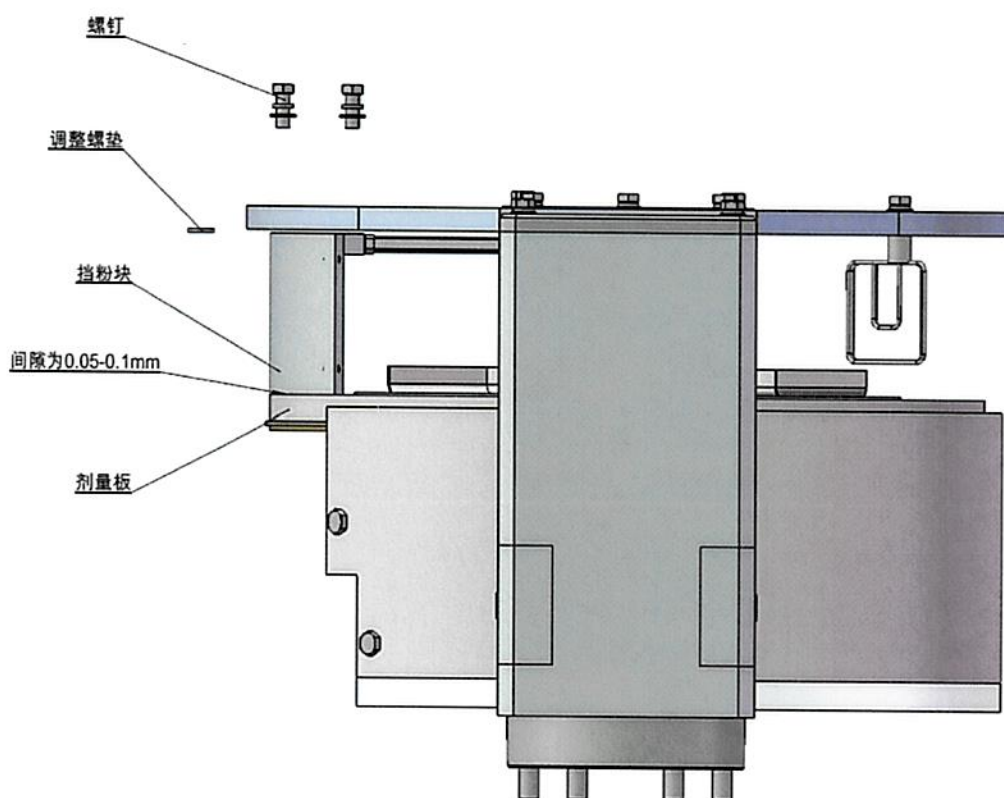
#### 1.4.7 Adjustment of Powder Stopper Clearance

挡粉器与剂量盘间隙为 0.05—0.1mm 范围。调整方法：松开螺钉，

The clearance between the powder stopper and the dose tray should be in the range of 0.05-0.1mm. Adjustment method: loosen the screws.

拆下垫片，以垫片厚度确定间隙，用塞尺来调整后，固禁螺钉。

Remove the gasket, determine the gap according to the thickness of the gasket, adjust with a feeler gauge, and then fix the screws tightly.



螺钉

screw

调整螺垫

adjusting shim

挡粉块

powder blocking piece

间隙为 0.05-0.1mm

gap is 0.05-0.1mm

剂量板

dosing plate

#### 1.4.8 粉位界限探测器的调整

#### 1.4.8 Adjustment of the powder level limit detector

电容式界限探测器用于控制粉箱内粉位高度。按粉流动性适当调节

Capacitive level detectors are used to control the powder level in the powder box. Adjust appropriately according to the flowability of the powder.

高度，探测器底部面距剂量盘上平面调整范围值为 18-28mm。

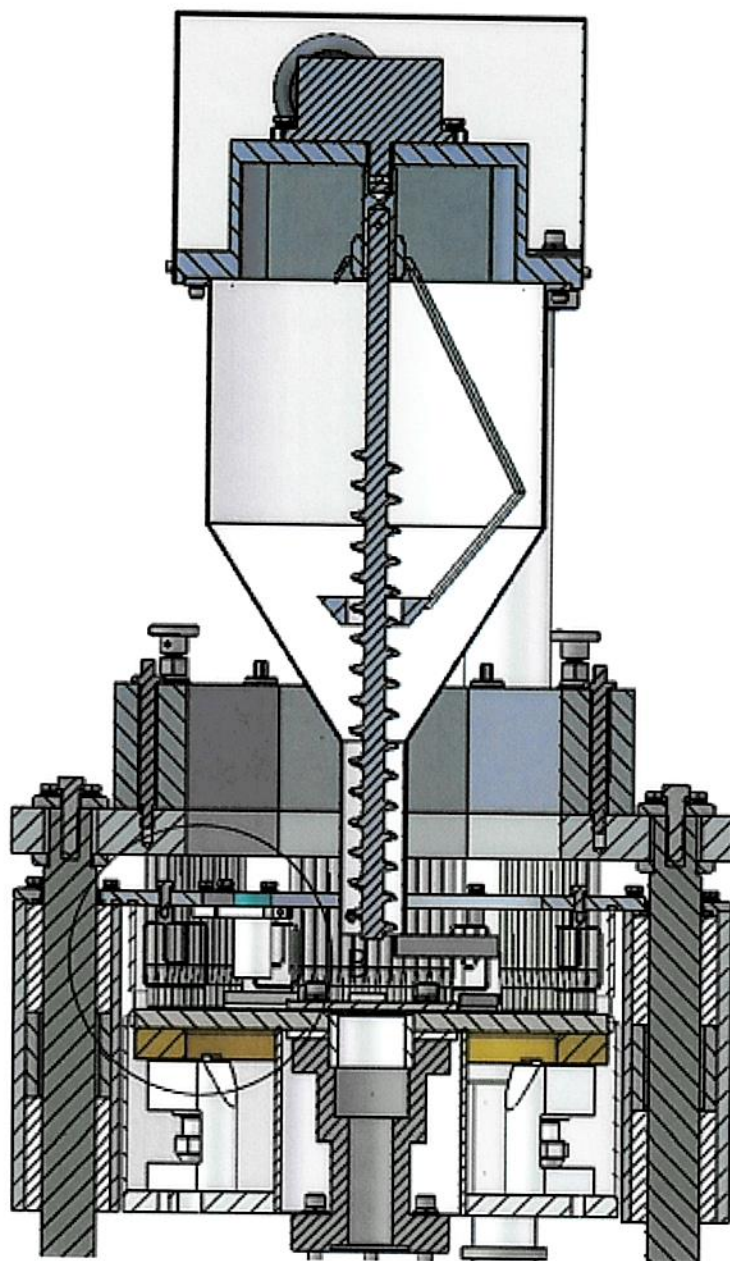
The adjustment range of the distance between the bottom surface of the detector and the plane on the dosing disk is 18-28mm.

调节方法：松开探测器支座上的 1 枚 M6 固紧螺钉，即可调整探测器高低位置。

Adjustment method: Loosen the 1 M6 fixing screw on the detector support to adjust the height position of the detector.

( 注意：探测器调整范围不该超出调整范围值 )

(Note: The adjustment range of the detector should not exceed the specified range.)



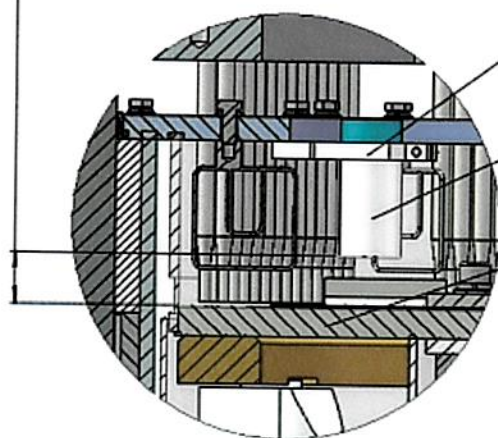


调整范围为18-28mm

检测器支架

检测器

剂量盘



调整范围为 18-28mm

Adjustment range is 18-28mm.

检测器支架

detector stand

检测器

detector

剂量盘

Dose plate

#### 1.4.9 冲杆座、冲杆高度调整

1.4.9 Punch rod seat, punch rod height adjustment.

剂量与粉柱密度：冲杆伸进剂量盘内的深度可调节。深度调节正

Dose and powder column density: The depth of the punch rod inserted into the dose tray can be adjusted. depth adjustment positive

确，粉量即准确。方法：转动主电机主轴，使冲杆部件下降到最低位置，调节冲杆端面与剂量盘同一平面，记下零点值（标尺数）后，按

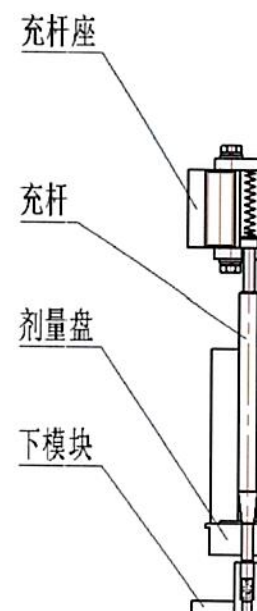
顺序把冲杆插入孔内，其深度按下表进行调节。

Yes, the powder quantity is accurate. Method: Rotate the main motor spindle to lower the punch rod component to the lowest position, adjust the end face of the punch rod to be in the same plane as the dose disk, record the zero point value (scale number), and then insert the punch rod into the hole in sequence, adjusting the depth according to the table below.

### 参考数值

reference numerical value

| 站<br>station               | 1 | 2 | 3 | 4 | 5   |
|----------------------------|---|---|---|---|-----|
| 插入深度<br>Insertion<br>depth | 6 | 4 | 3 | 2 | 0.5 |



### 充杆座

Charging handle base



充杆

Charging handle

剂量盘

Dose plate

下模块

Lower module

#### 1.4.10 报废胶囊剔除调整

#### 1.4.10 Scrap Capsule Removal Adjustment

该机械由顶针、把手、导柱、定位块、拉杆等零部件组成。

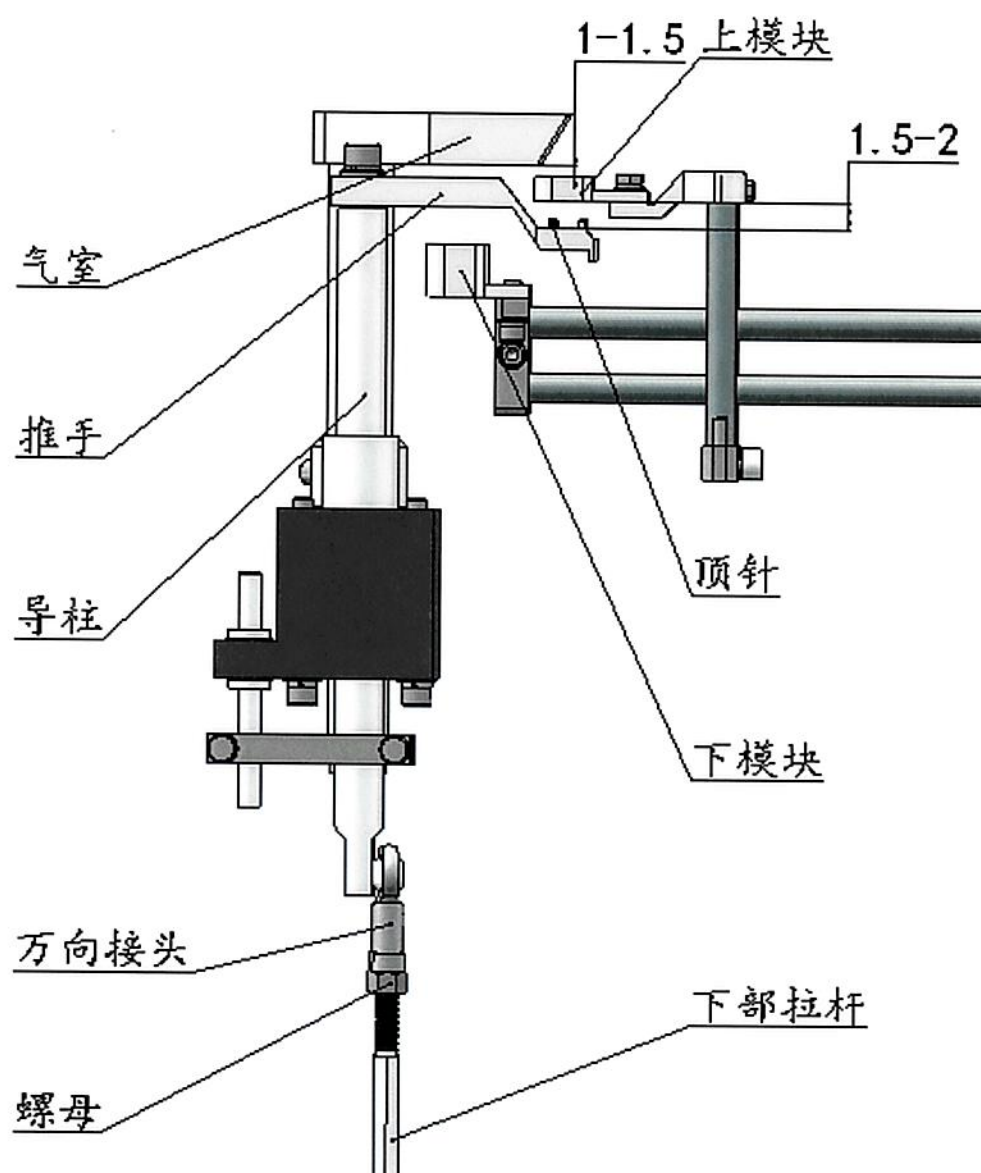
This machine is composed of components such as the top needle, handle, guide pillar, positioning block, and pull rod.

其功能是把帽、体未分离的废囊剔出，由吸气口吸入贮藏箱内。

Its function is to remove and discard the capsules that are not separated into cap and body, and suck them into the storage box through the suction port.

方法：调节拉杆，使把手平面与上模下平面间隙保留 1.5—2 mm 范围即可。调节气室，使气室下平面与上模块上平面间隙保留 1-1.5mm 范围即可。

Method: Adjust the pull rod to maintain a gap of 1.5-2mm between the handle plane and the lower plane of the upper mold. Adjust the air chamber to maintain a gap of 1-1.5mm between the lower plane of the air chamber and the upper plane of the upper mold.



气室

Air chamber

推力

Push handle

导柱

Guide pillar

万向接头

Universal joint

螺母

Nut

上模块

Upper module

顶针

Punch

下模块

Lower module

下部拉杆

Lower pull rod

#### 1.4.11 胶囊紧锁调整

#### 1.4.11 Capsule Tightening Adjustment

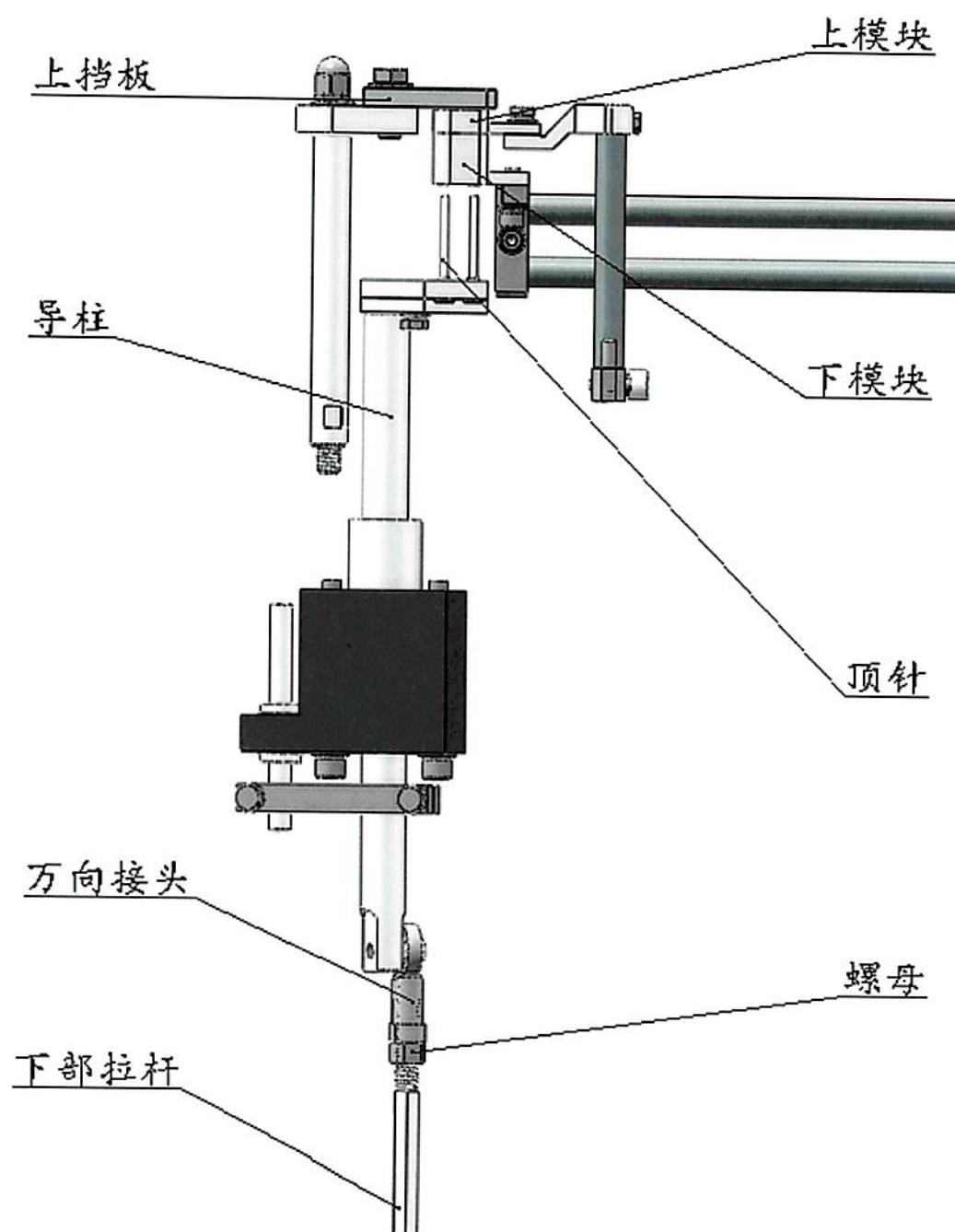
胶囊体完成填充后，到位帽体锁合。方法：转动拉杆，使顶针上升

After the capsule body is filled, the cap body is locked in place. Method:

Rotate the pull rod to make the top needle rise.

到离上档板距离等至于胶囊锁紧后的长度。（在后排孔内放入未锁胶囊，转到锁紧位观察测量长度校准为止）然后固紧螺钉、螺母。

The distance to the upper plate is equal to the length after the capsule is locked. (Insert the unlocked capsule into the front hole, turn it to the locked position, observe and measure the length for calibration) then tighten the screws and nuts.



上模块

Upper module

上挡板

Upper plate

导柱

Guide pillar

下模块

Lower module

顶针

Punch

万向接头

Universal joint

螺母

Nut

下部拉杆

Lower pull rod



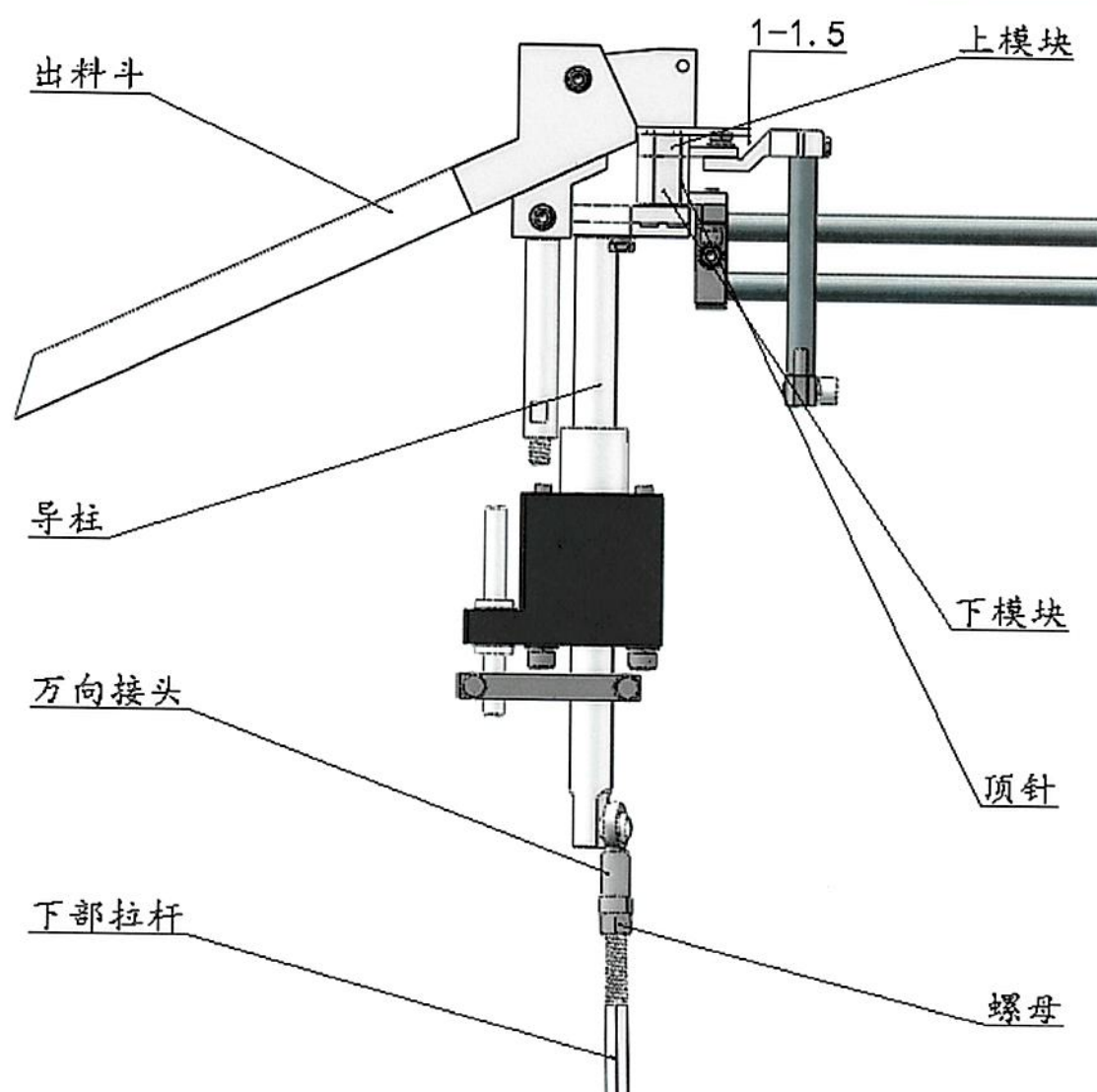
#### 1.4.12 出囊调节

#### 1.4.12 Capsule Adjustment

胶囊在前位锁紧后，已成成品，由该位推出。方法：(1) 转动拉杆，  
After the capsule is locked in the front position, it is considered a finished product and can be pushed out from that position. Method: (1) Rotate the pull rod.

使顶针上升，其端面高出上模平面 1—2mm，回程时，注意顶针端面与下模下平面保留间隙 1.5-2mm，防止碰撞。

Raise the needle bar so that the end face is 1-2mm above the upper die plane. During the return stroke, pay attention to maintaining a gap of 1.5-2mm between the needle end face and the lower die lower plane to prevent collision.



上模块

Upper module

上挡板

Upper plate

导柱

Guide pillar

下模块

Lower module

顶针

Punch

万向接头

Universal joint

螺母

Nut

下部拉杆

Lower pull rod

#### 1.4.13 传动凸轮调整

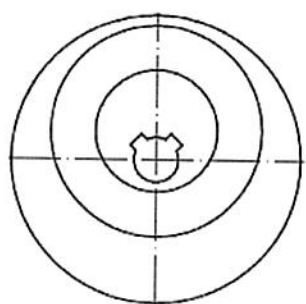
#### 1.4.13 Cam Adjustment

本机在装配时凸轮位置已调整好，切勿再进行调整，以防各部件

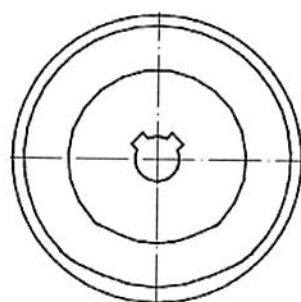
The cam position of this machine has been adjusted during assembly. Do not readjust it to prevent interference between components.

间发生干扰，如急需调整按钮仔细调整角度和位置。

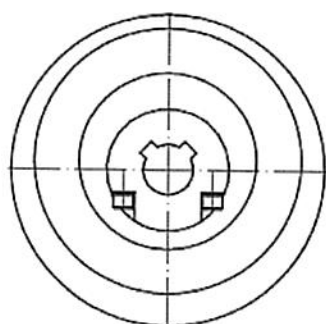
If adjustment is urgently needed, carefully adjust the angle and position of the knob.



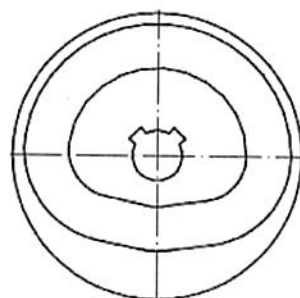
凸轮1 进料



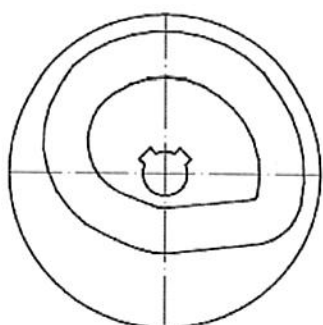
凸轮2 吸气



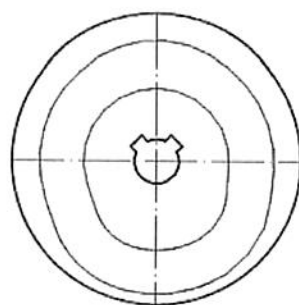
凸轮3 配料



凸轮4 出料



凸轮5 闭合



凸轮4 剔囊

凸轮 1 进料

Cam 1 Feeding

凸轮 2 吸气

Cam 2 sucks

凸轮 3 配料

Cam 3 Dosing

凸轮 4 出料

Cam 4 discharges

凸轮 5 闭合

Cam 5 closes

凸轮 4 剔囊

Cam 4 picking

#### 1.4.14 传动链调整

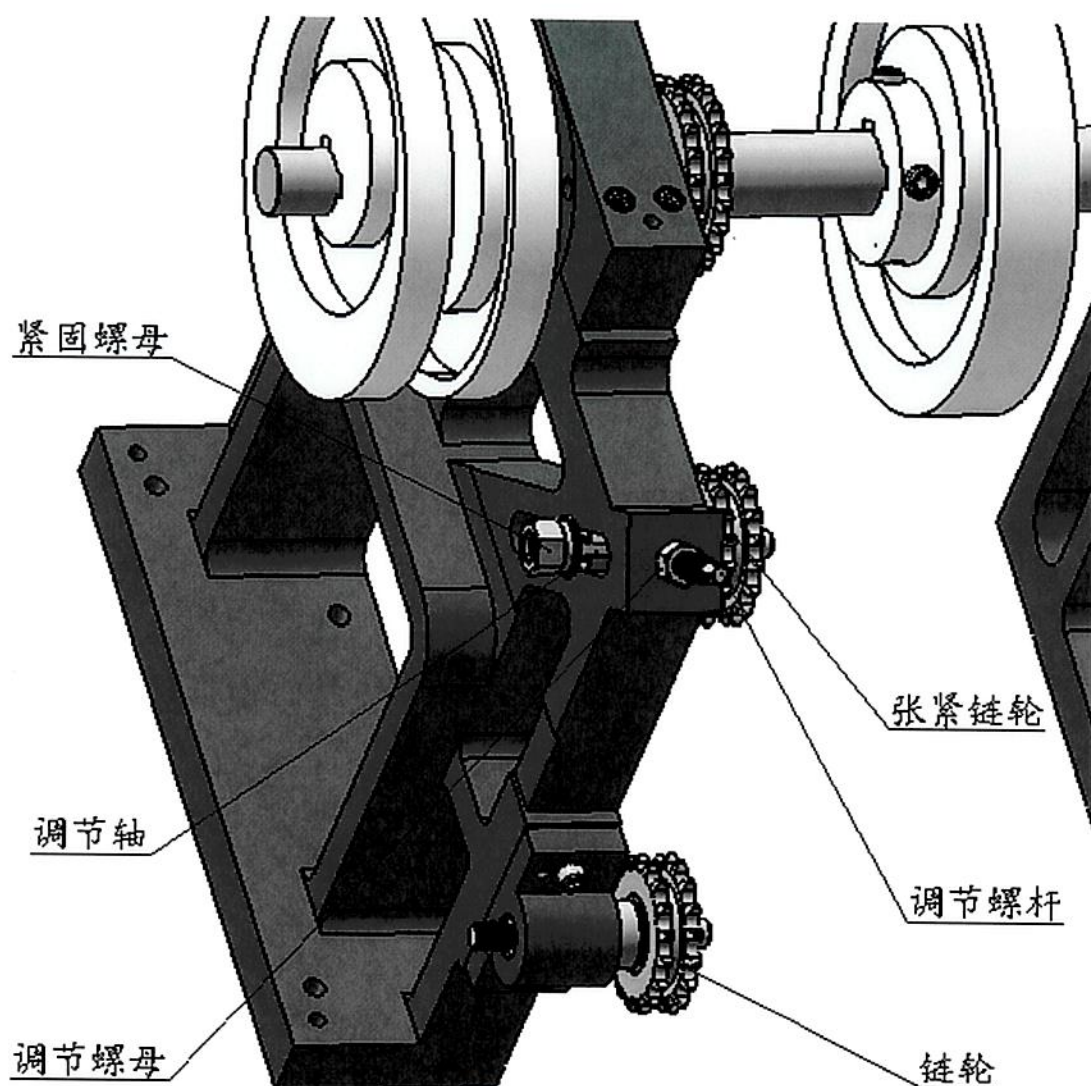
#### 1.4.14 Transmission Chain Adjustment

传动链由于使用时间长久产生拉长变松，影响传动精度。可改变

The transmission chain may become loose due to prolonged use, affecting transmission accuracy. Adjustable

张力轮位置，张紧后用螺母固紧。

Adjust the position of the tension wheel and tighten it with a nut after tensioning.





紧固螺母

Locking nut

张紧链轮

Tensioning sprocket

调节轴

Adjustment shaft

调节杆

Adjustment rod

调节母

Adjustment of the mother

链轮

sprocket

#### 1.4.15 真空度调整

#### 1.4.15 Vacuum Degree Adjustment

本机采用水环式 SK 系列真空泵。

This machine uses a water-ring SK series vacuum pump.

作用：播囊时能使胶囊准确入位，并能使胶囊帽体分离。

Function: During the capsule filling process, it can accurately position the capsule and separate the capsule cap and body.

真空度调整由水箱下部一只进水阀控制，度数由真空表上读取，一般在 0.02 — 0.04 Mpa 之间，确保胶囊正常分离，不会破损。

The adjustment of vacuum degree is controlled by a water inlet valve at the bottom of the water tank, and the degree is read from the vacuum gauge, generally between 0.02-0.04Mpa, to ensure the normal separation of the capsules without damage.



排气

Exhaust

排水

Drainage

吸气管路

Intake duct

排气管路

Exhaust duct

进水

Water inlet

1. 进水口

1. Water inlet port

11. 电磁阀

11. Solenoid valve

2. 积水放水口

2. Drain outlet for accumulated water

12. 旁通管路带截止阀

12. Bypass pipeline with stop valve

3. 溢流管

3. Overflow pipe

13. 过滤器(选用)

13. Filter (optional)

4. 排气管路

4. Exhaust pipe

14. 进水

14. Inlet water

5. 逆止阀

5. Check valve

15. 调节渗气阀(选用)

15. Adjust the air permeation valve (optional)

6. 过滤器(选用)

6. Filter (optional)

16. 气水分离器(选用)

16. Air-water separator (optional)

7. 控制阀

7. Control valve

17. 防气蚀孔(选用)

17. Anti-cavitation hole (optional)

8. 吸气管路

8. Suction pipe

18. 溢流口(选用)

18. Overflow port (optional)

9. 流量计

9. Flow meter

19.排气管路(选用)

19. Exhaust pipe (optional)

10.控制阀

Control valve

20.排水管路(选用)

20. Drainage pipe (optional)

#### 1.4.16 吸尘器调整

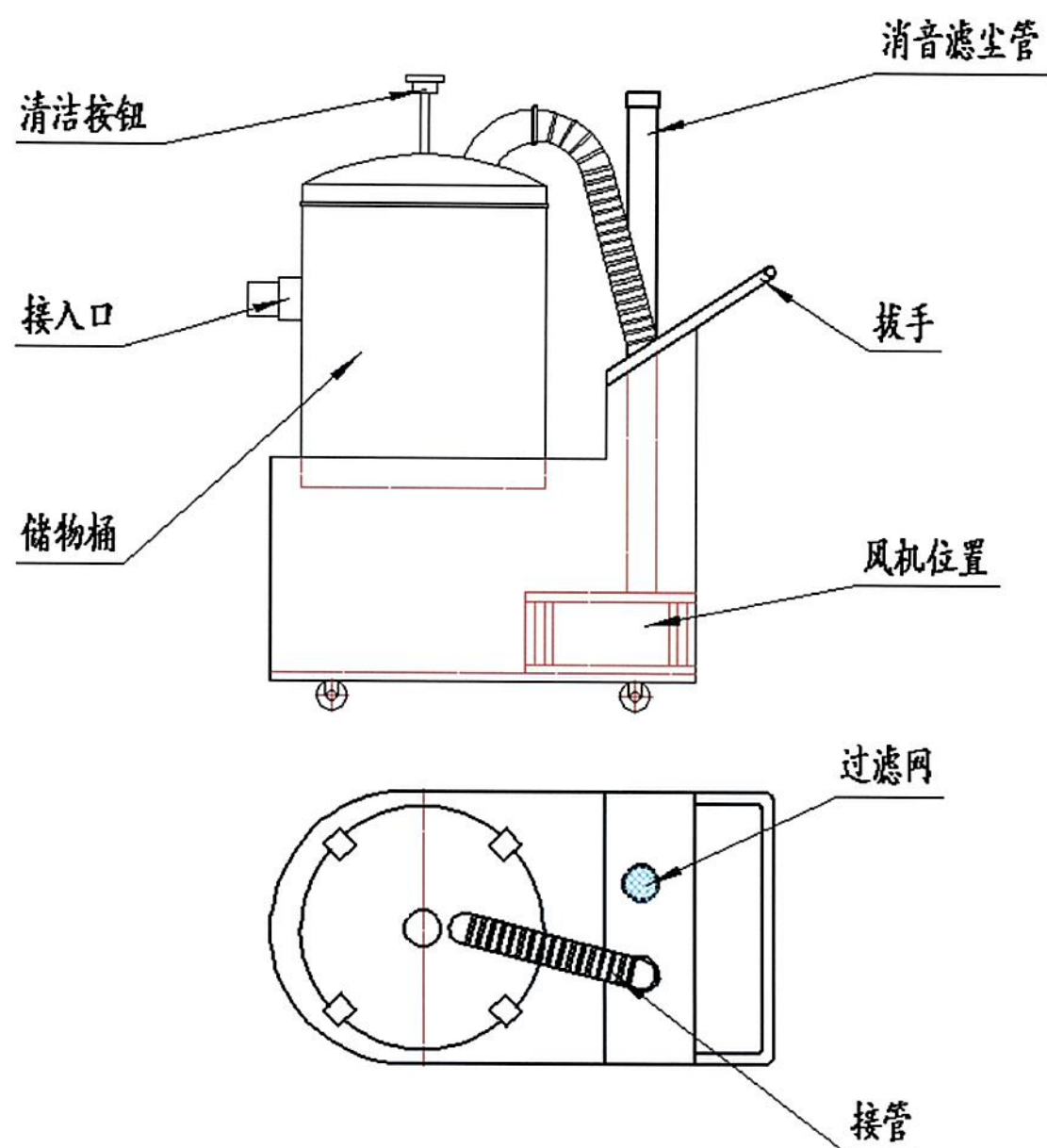
#### 1.4.16 Vacuum cleaner adjustment

吸尘器作用：把从各工位上粉尘，废胶囊吸入贮藏箱内。因工作

Vacuum cleaner function: suck the dust and waste capsules from each workstation into the storage box. Due to work

时间长，滤网布孔、吸尘管被阻塞，所以要定时拆下滤网刷洗干净，否则吸尘器会出现异常声音。

After a long time, the filter cloth holes and suction pipes may get blocked, so it is necessary to regularly remove the filter and clean it with a brush, otherwise the vacuum cleaner may produce abnormal noise.



消音滤尘管

Silencing and dust filtering pipe

清洁按钮

Clean button

接入口

Access port

拔手

Pull out the hand

储物桶

Storage punch

风机位置

Fan position

过滤网

Filter screen

接管

Joint pipe

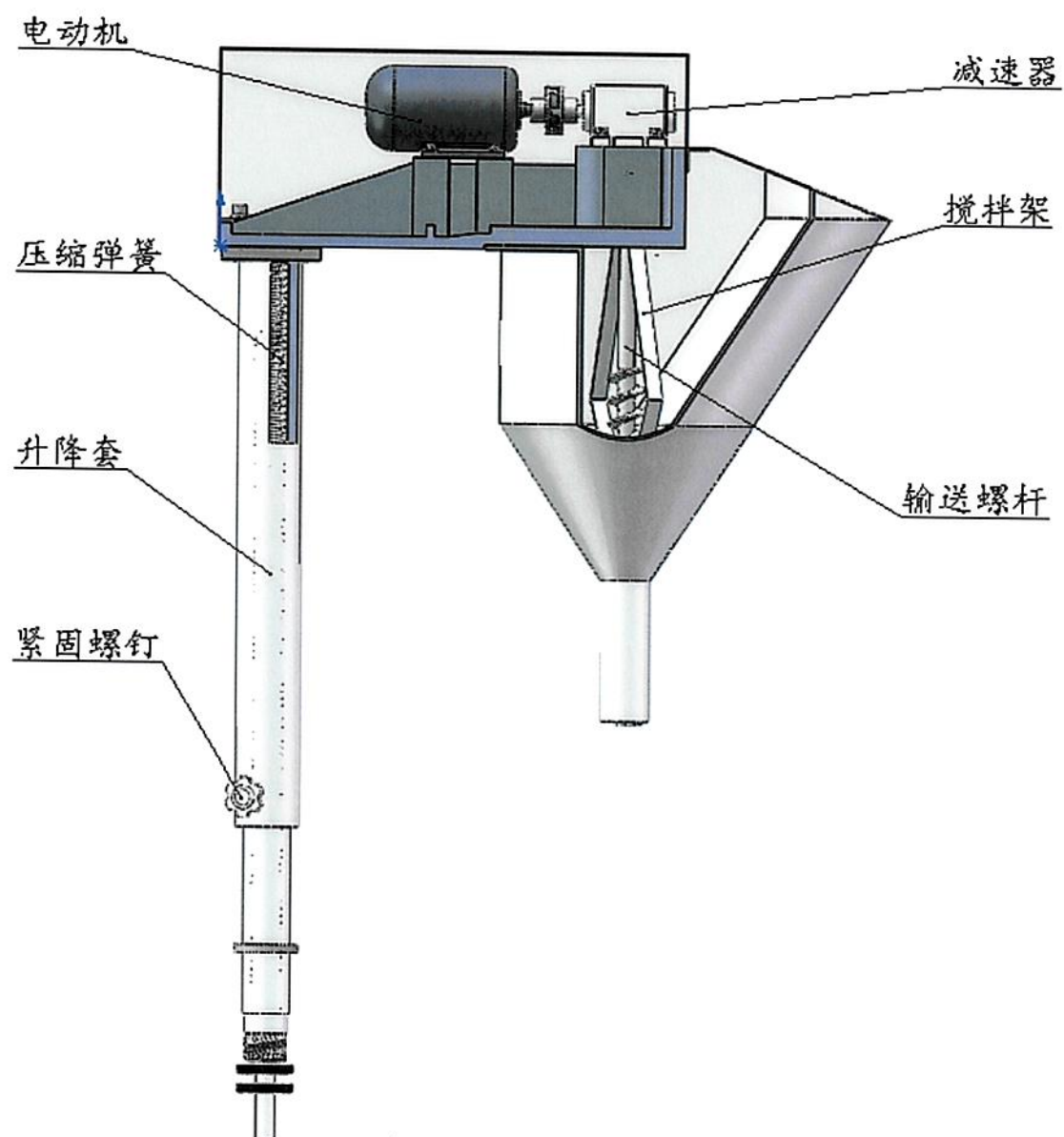


#### 1.4.17 粉末加料机构

#### 1.4.17 Powder Feeding Mechanism

粉末倒入料斗,搅拌架和输送螺杆转动使粉末均匀的进入填充位置。

Powder is poured into the hopper, and the stirring frame and conveying screw rotate to evenly feed the powder into the filling position.



电动机

Electric motor

压缩弹簧

Compression spring

升降套

Lift sleeve

紧固螺钉

Fastening screws

减速器

Reducer

搅拌架

Agitator frame

输送螺杆

Conveyor screw

## 1.5 机器的维护与清洗

## 1.5 Maintenance and Cleaning of the Machine

### 维修保养计划

#### Maintenance Plan

| 编号<br>Number | 维修保养点<br>Maintenance points      | 尺寸/数值概述<br>Dimension/Value<br>Overview |
|--------------|----------------------------------|----------------------------------------|
| 清洁           |                                  |                                        |
| 1<br>1.1     | 基本清洁<br>Basic cleaning<br>清洁机器内部 | 只能用吸尘器清洁<br>Use a vacuum cleaner only  |
| 2<br>2.1     | 日常清洁<br>Daily cleaning<br>清除残留包材 | 只能用吸尘器清洁<br>Use a vacuum cleaner only  |
| 3<br>3.1     | 传感器<br>sensor<br>清洁传感器           | 用柔软不起毛的布<br>Use a soft, non-lint cloth |

### 维修保养计划

#### Maintenance Plan

| 编号<br>Number    | 维修保养点<br>Maintenance points | 尺寸/数值概述<br>Dimension/Value<br>Overview |
|-----------------|-----------------------------|----------------------------------------|
| 维护保养            |                             |                                        |
| 4<br>4.1<br>4.2 | 齿轮箱<br>Gearbox<br>检测/装满润滑油  | 注意润滑油生产商信息<br>Attention to lubricant   |

| 5            |                                                                       |                                            |
|--------------|-----------------------------------------------------------------------|--------------------------------------------|
| 5.1          | 检查电器设备是否有损坏<br>Check if there is any damage                           | 关掉开关并互锁！<br>Turn off the switch and        |
| 6            |                                                                       |                                            |
| 6.1          | 检查门触开关是否有损坏，<br>确保正常工作                                                |                                            |
| 7            |                                                                       |                                            |
| 7.1          | PLC<br>PLC<br>PLC 每隔 12 个月更换一次                                        |                                            |
| 8            |                                                                       |                                            |
| 8.1          | 检查紧固件，确保稳固连接<br>Check fasteners to ensure                             | 及时更换松动的自锁螺丝<br>Renlace loose self-locking  |
| 9            |                                                                       |                                            |
| 9.1          | 检查计时带是否有损坏，确<br>保张紧                                                   |                                            |
| 10           | 组件<br>Components                                                      |                                            |
| 10.1         |                                                                       | 打开之前先减压<br>Release pressure before opening |
| 10.2         | 排空过滤器罩                                                                |                                            |
| 11           | 真空系统<br>Vacuum system                                                 | 包装精细的，不可压缩的产品<br>时须定期清洁，如 OTC 产品<br>或食品添加剂 |
| 11.1         | 清洁过滤装置                                                                |                                            |
| 11.2         |                                                                       |                                            |
| 11.3         | Clean the filter device                                               |                                            |
| 12           | 径向凸轮<br>Radial cam                                                    | STABYL 300 AL 2<br>STABYL300AL2            |
| 12.1         | 润滑                                                                    |                                            |
| 13           | 定向装置<br>Directional device                                            | 1-2strokes ,Fuchs GERALYN<br>1             |
| 13.1         | 导轨，定向推进器，各有一个润<br>滑喷嘴                                                 | 1-2strokes ,Fuchs GERALYN<br>1             |
| 13.2         |                                                                       |                                            |
| 13.3         | Guide rails, directional thrusters,<br>each with a lubrication nozzle | STABYL 300 AL 2                            |
| 13.4         |                                                                       |                                            |
| 编号<br>Number | 维修保养点<br>Maintenance points                                           | 尺寸/数值概述<br>Dimension/Value<br>Overview     |
| 维护保养         |                                                                       |                                            |
| 14           | 微丸充填工位<br>Microsphere filling station                                 | STABYL 300 AL 2<br>STABYL300AL2            |
| 14.1         | 润滑凸轮轨道和填充杆                                                            |                                            |
| 15           | 废胶囊剔除工位<br>Waste capsule removal<br>station                           | STABYL 300 AL 2<br>STABYL300AL2            |
| 15.1         |                                                                       |                                            |

|            |                                               |                                 |
|------------|-----------------------------------------------|---------------------------------|
| 16<br>16.1 | 胶囊闭合工位<br>Capsule closing station<br>润滑扭转锁    | STABYL 300 AL 2<br>STABYL300AL2 |
| 17<br>17.1 | 胶囊出货工位<br>Capsule dispensing station<br>润滑扭转锁 | STABYL 300 AL 2<br>STABYL300AL2 |

维  
修  
保  
养  
计  
划

## Maintenance Plan

## 1.6 零件及模具的拆卸与更换

## 1.6 Disassembly and Replacement of Parts and Molds

模具的更换有上下模具、填充杆、剂量盘、播囊管部分、胶囊梳等。

Mold replacement includes upper and lower molds, filling rods, dosing discs, capsule tube parts, capsule combs, etc.



选购产品：微丸灌装装置

Product selection: Microgranule filling device

#### 1.6.1 更换拆卸上、下模具及密封圈

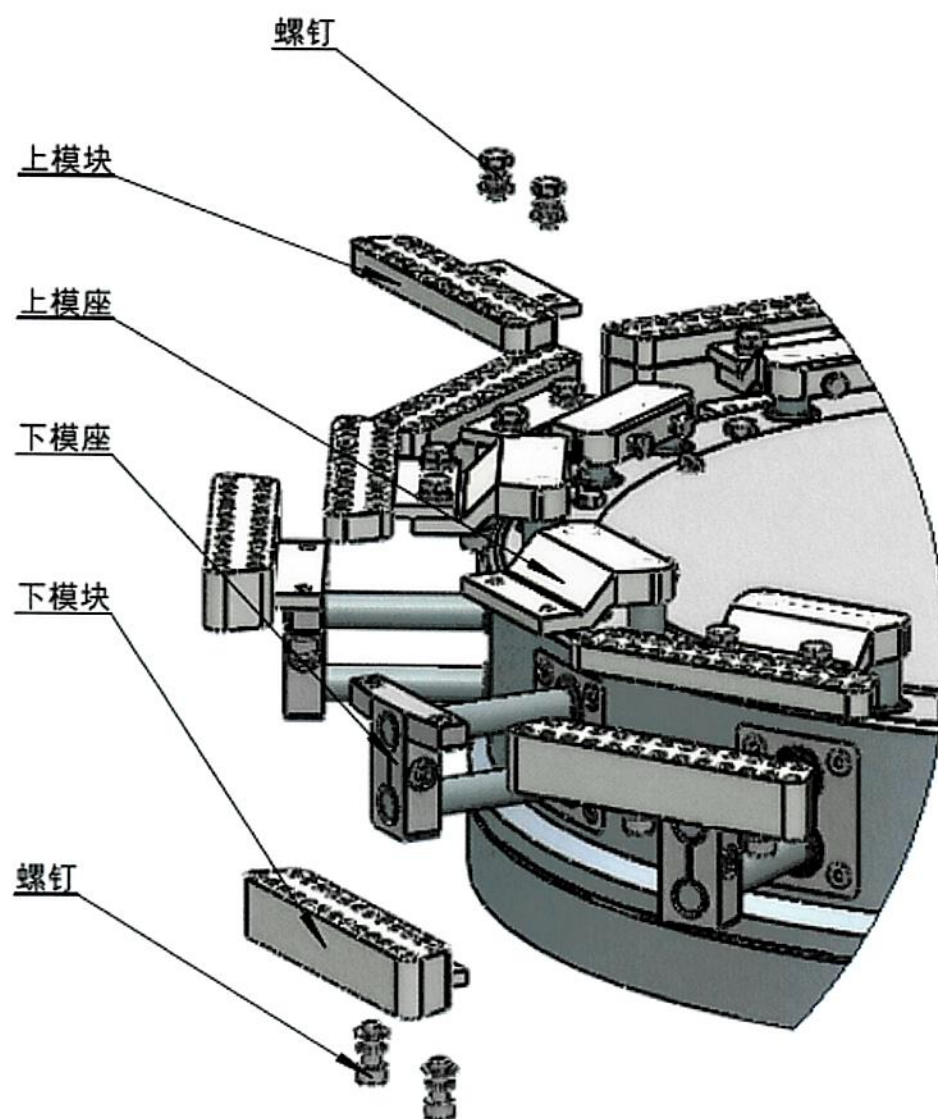
1.6.1 Replacement and disassembly of upper and lower molds and sealing rings

a. 松开工位转盘上模座与下模座的螺钉，逐一取下上下模具即可。  
松开内六角螺栓并取出下模座，再松开四枚螺钉拿出压板和密封圈，  
换上新密封圈依次安装。

Loosen the screws of the upper and lower mold seats on the workstation turntable, and then remove the upper and lower molds one by one.  
Loosen the internal hexagon bolt and remove the lower mold base, then loosen the four screws to take out the pressure plate and sealing ring, replace with a new sealing ring and install them in order.

（注意：安装、换取时应尽量避免模具磕碰）

(Note: Try to avoid bumping the mold during installation and replacement)



螺钉

screw

上模块

Upper module

上模座

Upper mold base

下模座

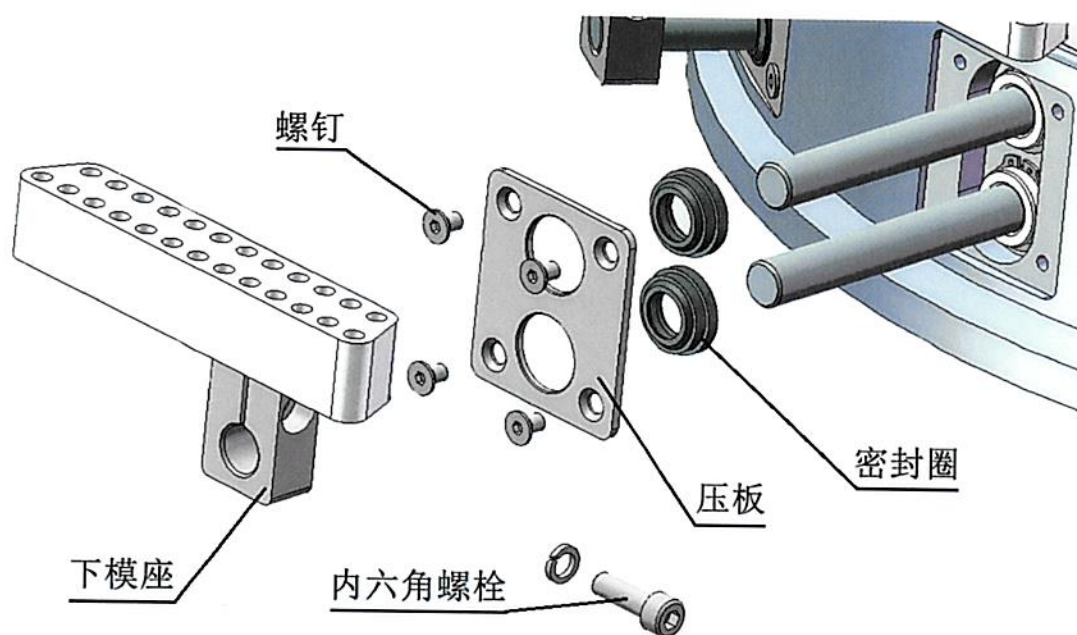
Lower mold base

下模块

Lower module

螺钉

screw



螺钉

screw

密封圈

Sealing ring

压板

platen

下模座

Lower mold base



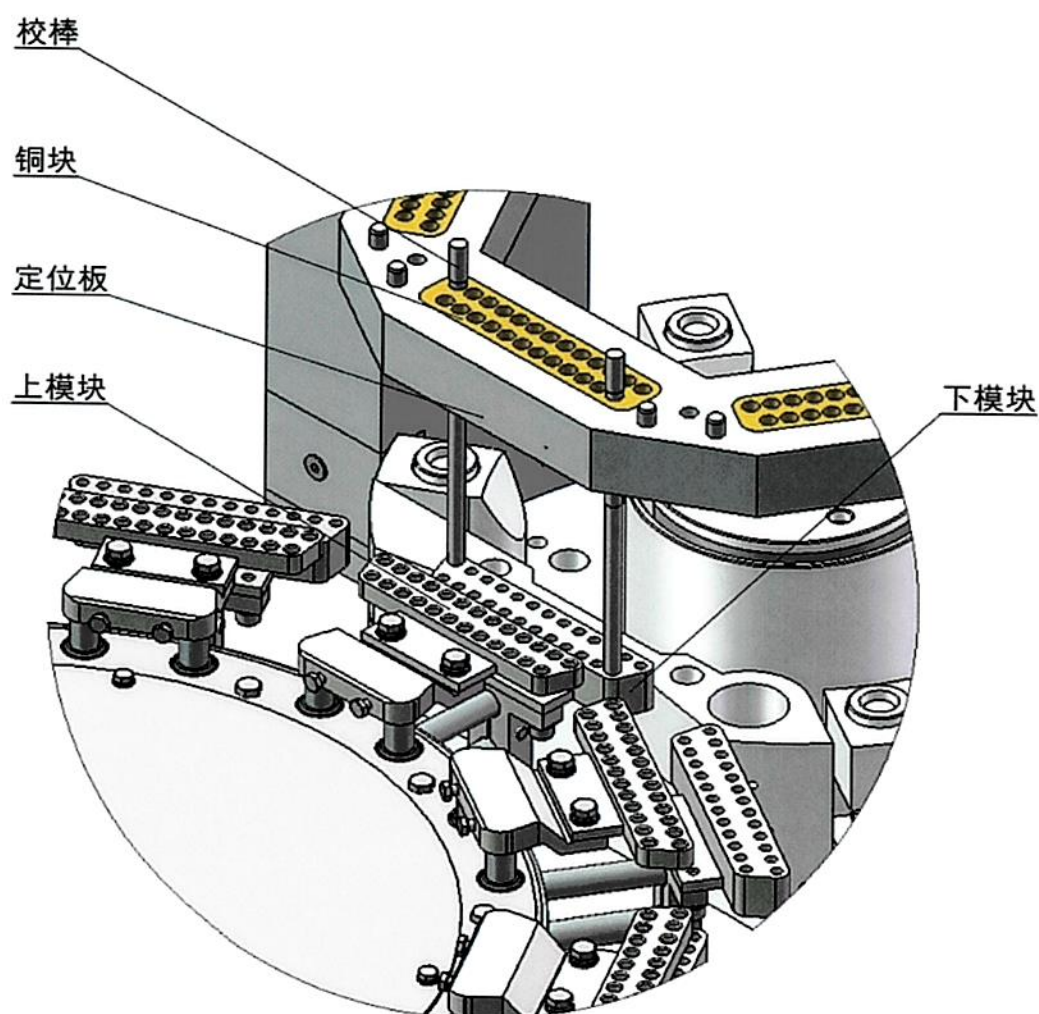
## 内六角螺栓

socket bolts

b.将替换模具对位安装，另拆卸粉料填充部分的填充模块，以定位板前排铜块上的充杆定位孔为基准，取出附件工具中2条与下模块模号相对应的调试校棒；分别将2条校棒垂直插入充杆定位孔（如图所示）；利用校棒逐孔校准，当下模块上模孔与定位板前排铜块上的充杆定位孔同心后；打开传动部分的侧门板，取出附件工具中的手轮，套入主电机输入轴，（顺时针）手动转动电机输入轴，并观察验证校棒与下模块模孔是否已经对准（不得出现异响或擦皮现象），空转3—5周，确认定位准确后取出校棒。

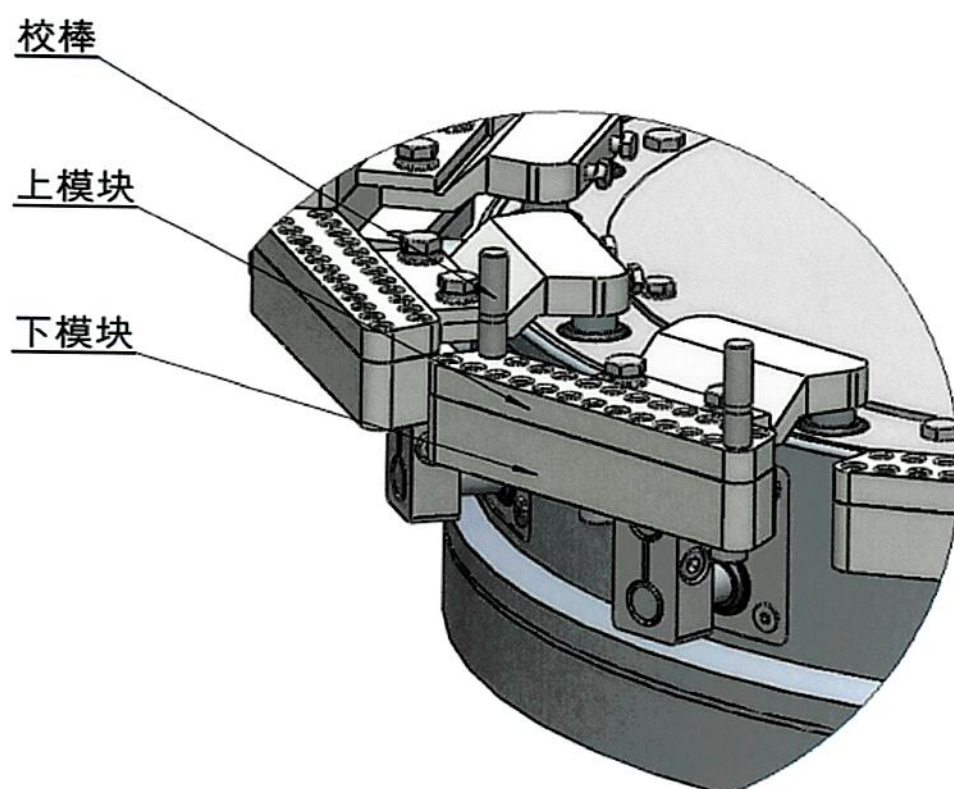
Install the replacement mold in alignment, and disassemble the filling module of the powder filling part. Taking the filling rod positioning holes on the copper block in the front row of the positioning plate as a benchmark, take out the 2 debugging calibration rods corresponding to the mold number of the lower module in the accessory tool; Insert the 2 calibration rods vertically into the filling rod positioning holes respectively (as shown in the figure); calibrate them one by one with the calibration rods. After the upper mold holes of the lower module are concentric with the filling rod positioning holes on the copper block in the front row of the positioning plate; Open the side door plate of the transmission part, take out the handwheel in the accessory tool, put it on

the main motor input shaft, manually turn the motor input shaft (clockwise), and observe and verify whether the calibration bar is aligned with the mold hole of the lower module (no abnormal noise or peeling), idle for 3-5 weeks, and take out the calibration bar after confirming that the positioning is accurate.



c.接着以下模块为基准，取出附件工具中的对应校棒，在锁紧、出囊、播囊等工位利用类似方法逐孔校准上模块。

Then the following modules are used as a benchmark, take out the corresponding calibration rods in the accessory tools, and calibrate the upper modules hole by hole at locking, bag extraction, bag sowing and other stations using a similar method.



校棒

Alignment pin

上模块

Upper module

下模块

Lower module

#### 1.6.2.1 清洗拆卸粉末填充模块

#### 1.6.2.1 Cleaning and dismantling the powder filling module

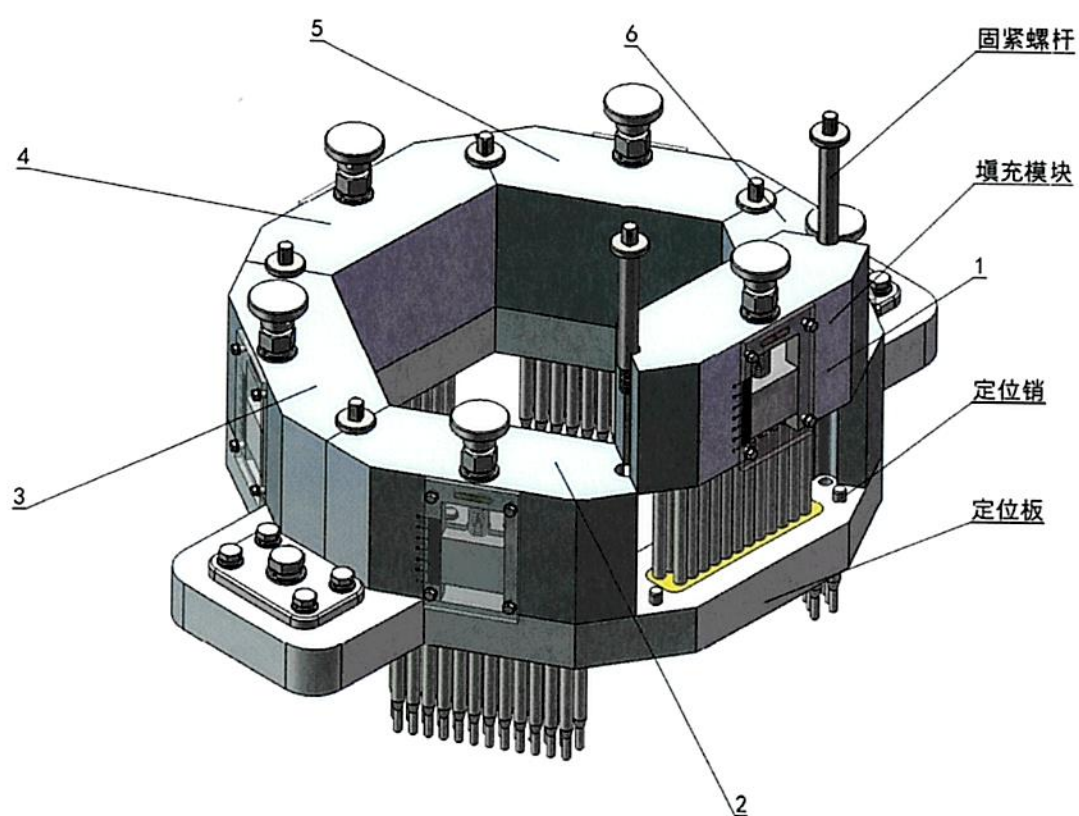
松开固紧螺杆，握紧填充模块的手轮或者充杆座支撑套，将模块从定位板垂直向上取出，清洗完毕后将模块重新放入并固紧。

Loosen the locking screws, grip the handwheel or support sleeve of the filling module tightly, and lift the module vertically upward from the positioning plate. After cleaning, reinsert the module and tighten it.

（注意：充杆座支撑套底面刻有该支撑套的编号，它将与定位板上六个方位刻着的编号相对应，安装调试时请对号入座）

(Note: The bottom of the support sleeve is marked with the corresponding number, which will match the six numbers marked on the positioning plate. Please align them during installation and debugging.)





固紧螺纤

Fastening screw fiber

填充模块

Filling module

定位销

Positioning pin

定位板

Positioning plate

#### 1.6.2.2 更换拆卸粉末填充模块的填充杆及压缩弹簧

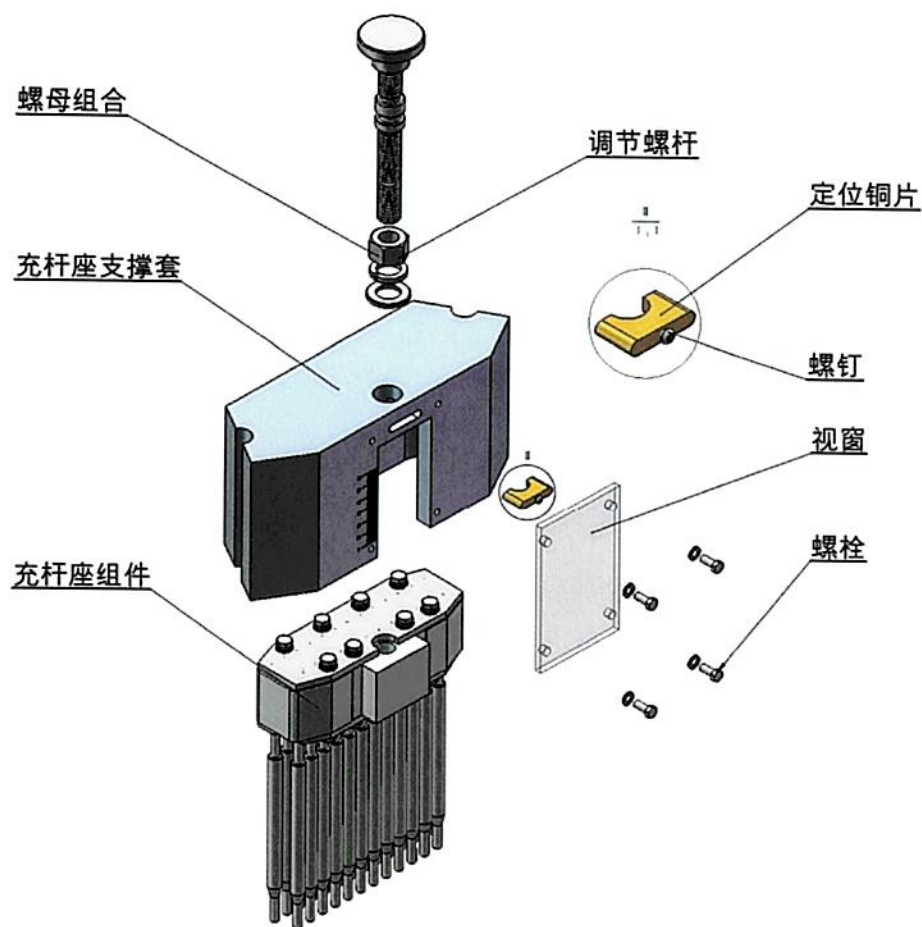
### 1.6.2.2 Replacement and disassembly of the filling rod and compression spring of the powder filling module

a、松开视窗上的四枚紧固螺钉，将附件工具中的一枚 M3 小螺钉拧入定位铜片的螺纹孔中，小螺钉充当小把手，将定位铜片从充杆座支撑套中取出；松开充杆座支撑套顶部的螺母，退出调节螺栓，取出充杆座组件。

a. Loosen the four fastening screws on the window, screw an M3 small screw from the accessory tool into the threaded hole of the positioning copper piece, use the small screw as a handle to remove the positioning copper piece from the support sleeve. Loosen the nut at the top of the support sleeve, remove the adjusting bolt, and take out the support sleeve assembly.

( 注意：1.此时充杆座组件，失去固定，可能会滑出充杆座支撑套 2.充杆座侧面刻有该充杆座的编号，它将与充杆座支撑套底部的编号相对应，安装调试时请对号入座 )

(Note: 1. At this point, the support sleeve assembly is not fixed and may slide out of the support sleeve. 2. The side of the support sleeve is marked with the corresponding number, which should align with the number on the bottom of the support sleeve during installation and debugging.)



螺母组合

Nut assembly

调节螺杆

Adjusting screw

定位铜片

Positioning copper sheet

充杆座支撑套

Charging rod seat support kit

螺钉

screw



视窗

Window

螺栓

Bolt

充杆座组件

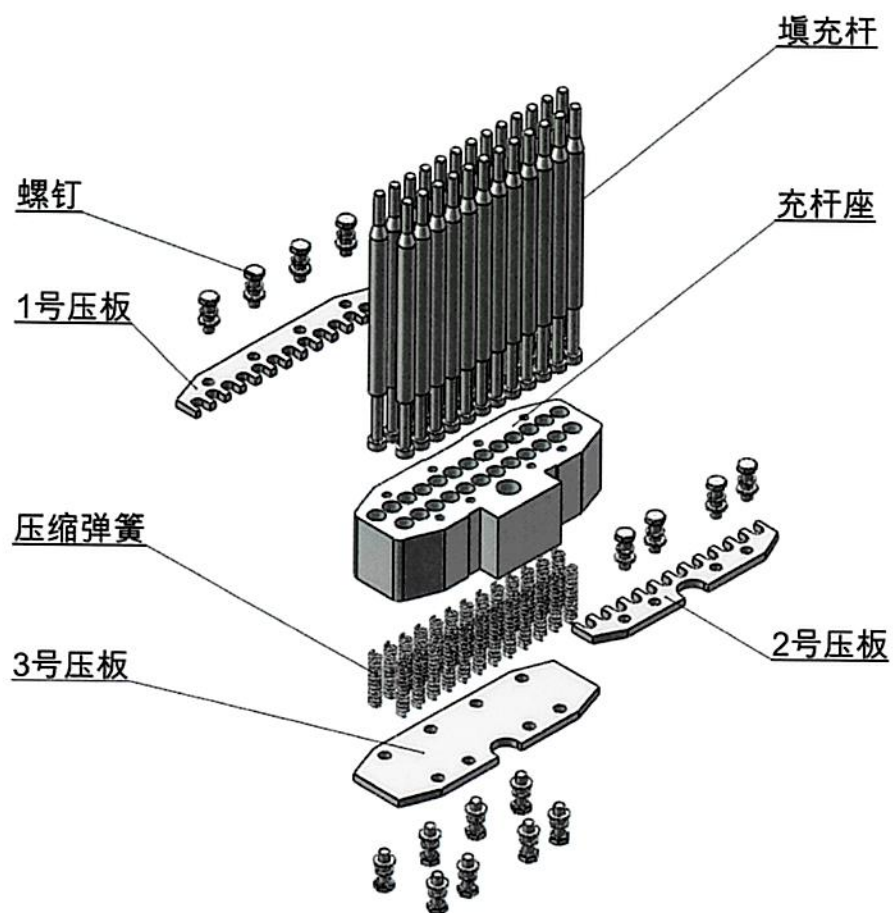
Filling rod seat assembly

b、松开充杆座组件 1 号、2 号及 3 号压板上的螺钉，取出压缩弹簧和填充杆进行更换。

b. Loosen the screws on the 1st, 2nd, and 3rd pressure plates of the charging rod seat assembly, and remove the compression spring and charging rod for replacement.

(注意：1.填充杆可能掉出 2.1 号填充模块和其他 5 个填充模块不同，只有 1 号压板和 2 号压板，充杆座内并无压缩弹簧)

(Note: 1. The charging rod may fall out. 2. The 1st filling module is different from the other 5 filling modules, with only the 1st pressure plate and 2nd pressure plate, and there is no compression spring in the charging rod seat.)



填充杆

Filling rod

螺钉

screw

充杆座

Charging handle base

1 号压板

Plate 1

压缩弹簧

Compression spring

2 号压板

Plate 2

3 号压板

Plate 3

### 1.6.2.3 清洗拆卸填充定位板

### 1.6.2.3 Cleaning and disassembling the filling positioning plate

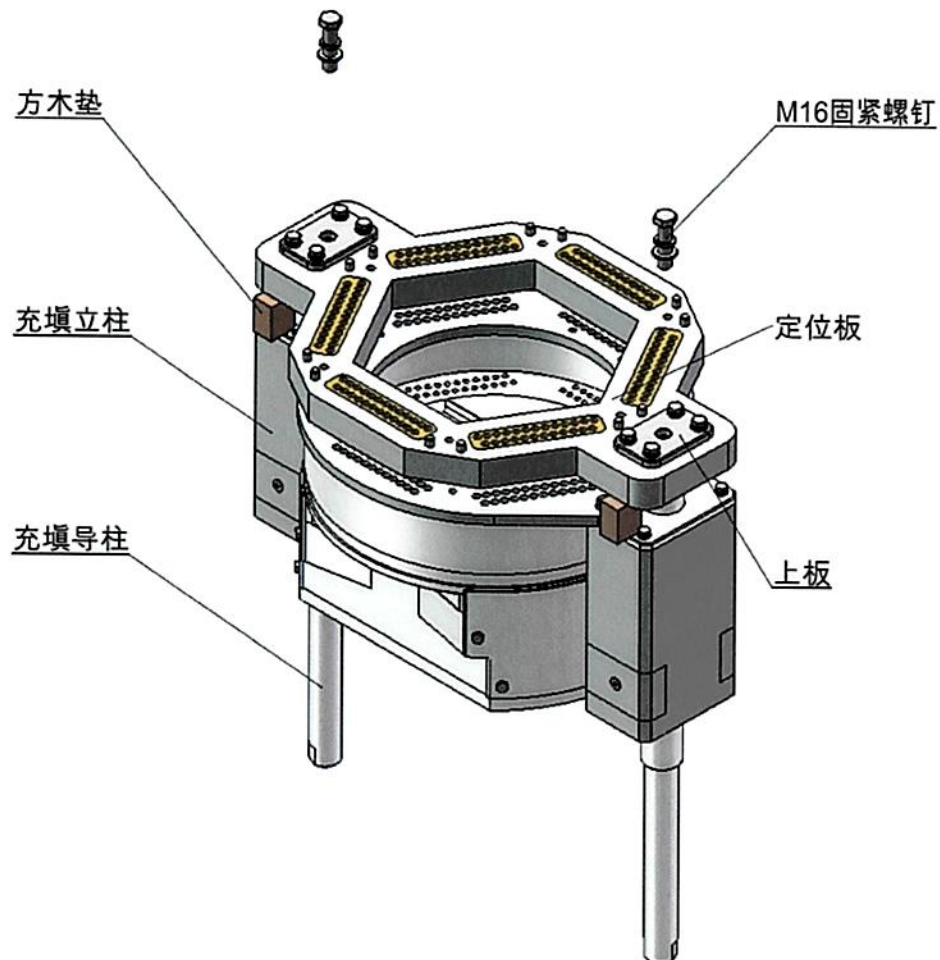
打开传动部分的侧门板，取出附件工具中的手轮，套入主电机输入轴，（顺时针）手动转动电机输入轴，使定位板上升到顶部。松开上板中间位置的 2 只 M16 紧固螺栓，用 2 块等厚木块垫入定位板下平面与充填立柱上平面之间，再次用手（顺时针）转动电机输入轴，使 2 条导柱回程，取下定位板。

Open the side door panel of the transmission section, take out the handwheel from the accessory tools, fit it into the main motor input shaft, (clockwise) manually rotate the motor input shaft to raise the positioning plate to the top. Loosen the 2 M16 tightening bolts in the middle position of the upper plate, place 2 equally thick wooden blocks between the lower surface of the positioning plate and the upper surface of the filling column, and then rotate the motor input shaft by hand (clockwise) to make the 2 guide pillars return, and remove the positioning plate.

（注意，此操作需要两人配合，导柱回程后，定位板失去限位，应避免其滑落）

(Note: This operation requires two people to cooperate. After the guide

pillars return, the positioning plate loses its positioning, so care should be taken to prevent it from slipping off)



方木垫

Square wooden pad

M16 固紧螺钉

M16 tightening bolt

充填立柱

Filling column

-定位板

Positioning plate



充填导柱

Filling guide pillar

.上板

.Upper plate

### 1.6.3 更换拆卸粉末剂量盘

#### 1.6.3 Replacement and disassembly of powder dosing disc

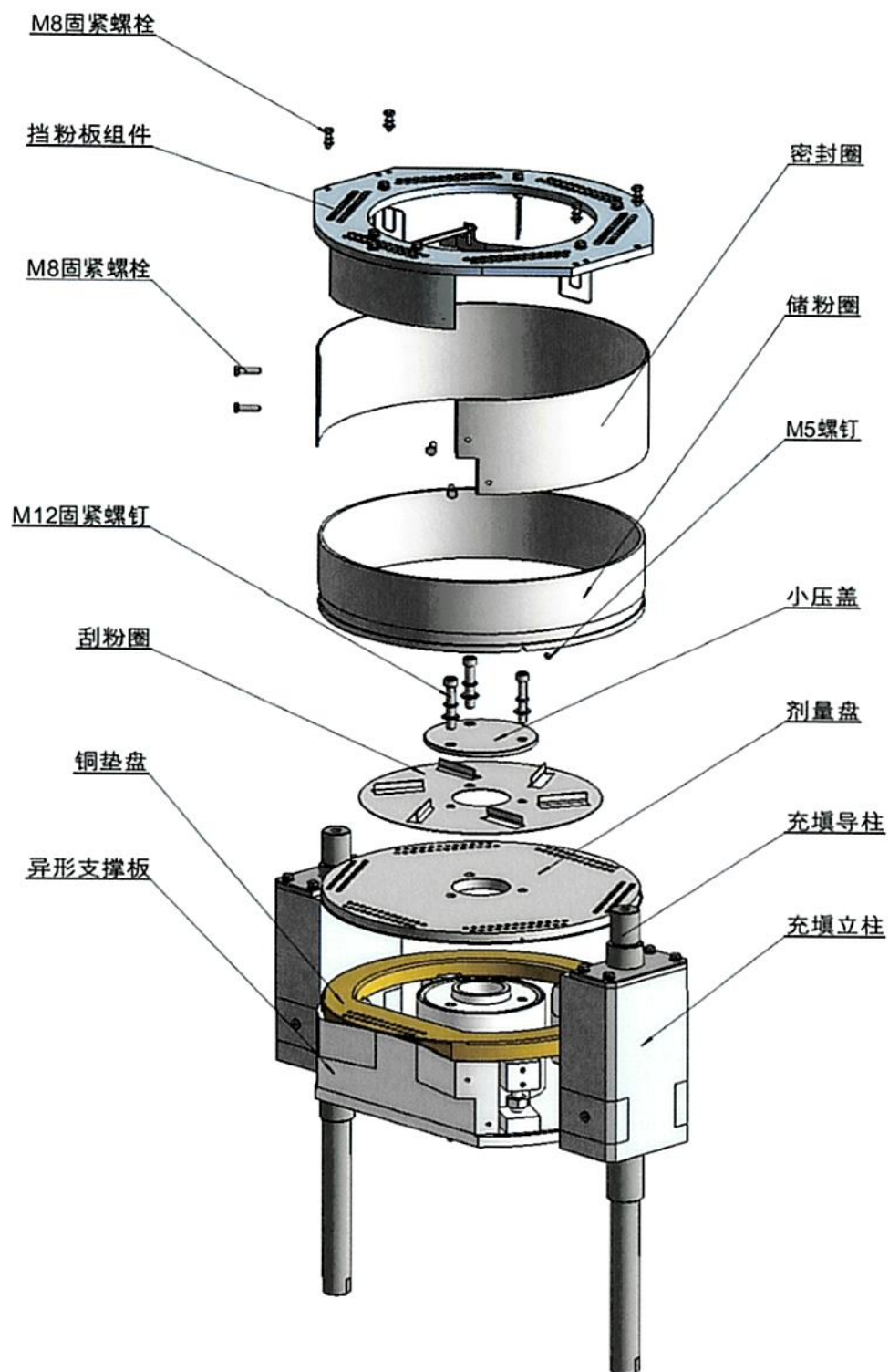
A.拆卸更换 松掉挡粉板组件与充填立柱连接的 4 枚 M8 固紧螺栓，取下挡粉板组件；松掉密封圈外侧与异形支撑板连接的 4 枚 M8 固紧螺栓，取出密封圈，露出储粉圈外侧的 3 枚 M5 螺钉；将 M5 螺钉松掉，取下贮粉圈；接着将小压盖上 3 枚 M12 内六角螺钉松开，取下小压盖、刮料圈和剂量盘。

Disassembly and replacement: Loosen the 4 M8 tightening bolts connecting the powder blocking plate assembly to the filling column, and remove the powder blocking plate assembly; loosen the 4 M8 tightening bolts connecting the outer side of the sealing ring to the shaped support plate, remove the sealing ring, exposing the 3 M5 screws on the outer side of the powder storage ring; loosen the M5 screws, remove the powder storage ring; then loosen the 3 M12 internal hexagon screws on the small pressure cover, remove the small pressure cover, scraping ring, and dosing disc.

( 注意：安装、换取时应尽量避免剂量盘磕碰 )

(Note: Avoid bumping the dose plate during installation and

replacement)



M8 固紧螺栓

Tighten the M8 bolts

挡粉板组件

Powder blocking plate assembly

M8 固紧螺栓

Tighten the M8 bolts

M12 固紧螺钉

Tighten the M12 screws

刮粉圈

Scraping ring

铜垫盘

Copper gasket

异形支撑板

Alien support plate

密封圈

Sealing ring

储粉圈

Powder storage ring

M5 螺钉

M5 screw

小压羔

Small pressure lamb

剂量盘



Dose plate

充填导柱

Filling guide pillar

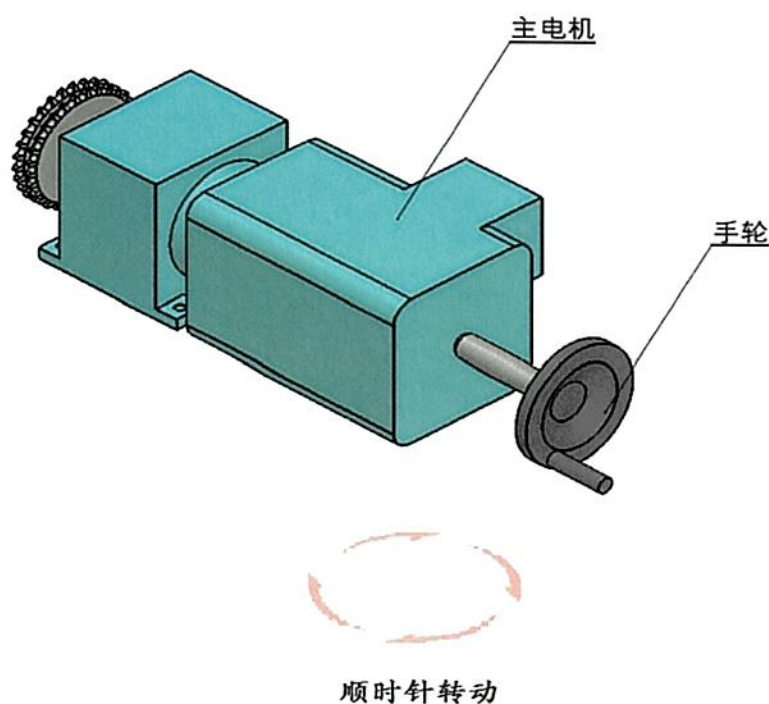
充填立柱

Filling column

B.安装定位：将替换的剂量盘装于法兰盘上，将定位板装入充填导柱并拧紧相关螺栓，以定位板铜块上的充杆定位孔为基准，将填充模块逐个放入定位板，利用填充模块上的填充杆校准剂量盘。当填充杆与剂量盘模孔定位同心后，按序将刮粉圈和小压盖装于剂量盘上，并固紧剂量盘上的 3 枚 M12 螺钉。打开传动部分的侧门板，取附件工具中的手轮套入主电机输入轴，（顺时针）手动转动电机输入轴，并观察充杆与剂量盘孔是否对准，逐排校正，不得出现擦皮现象，空转 3—5 周，确认定位准确后。重新拆下填充模块和定位板，然后顺序装上储粉圈、密封圈、挡粉板组件、定位板及填充模块，固紧相关紧固件。（注意：安装、换取时应尽量避免剂量盘磕碰）

B. Installation positioning: Place the replacement dose plate on the flange plate, install the positioning plate into the filling guide column and tighten the relevant bolts. Use the positioning hole on the copper block of the positioning plate as a reference, place the filling modules one by one on the positioning plate, and calibrate the dose plate using the filling rod on the filling module. When the filling rod is concentric with the dose plate module hole, sequentially install the scraper ring and small pressure

cover on the dose plate, and tighten the 3 M12 screws on the dose plate. Open the side door of the transmission section, take the handwheel from the accessory tools and fit it into the main motor input shaft, manually rotate the motor input shaft (clockwise), and observe whether the charging rod aligns with the dose disk hole. Correct them one by one, without any scraping, rotate the shaft empty for 3-5 rounds, and confirm the accurate positioning. Remove the filling module and positioning plate, then sequentially install the powder storage ring, sealing ring, powder blocking plate assembly, positioning plate, and filling module, and tighten the relevant fasteners. (Note: Avoid bumping the dose plate during installation and replacement)



主电机

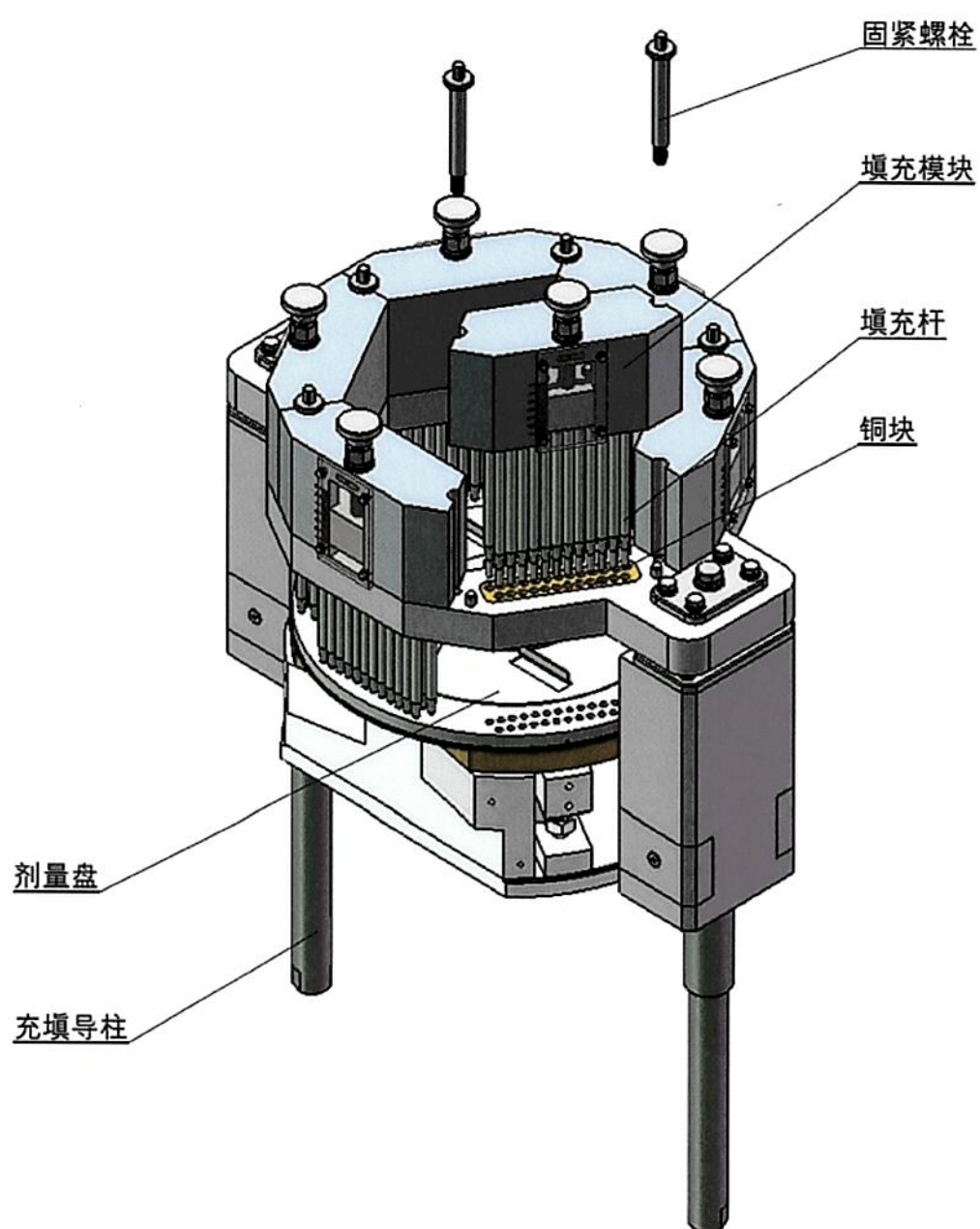
Main motor

手轮

Handwheel

顺时针转动

Clockwise rotation



固紧螺栓

Tighten the bolts

填充模块

Filling module

填充杆

Filling rod

铜块

Copper block

剂量盘

Dose plate

充填导柱

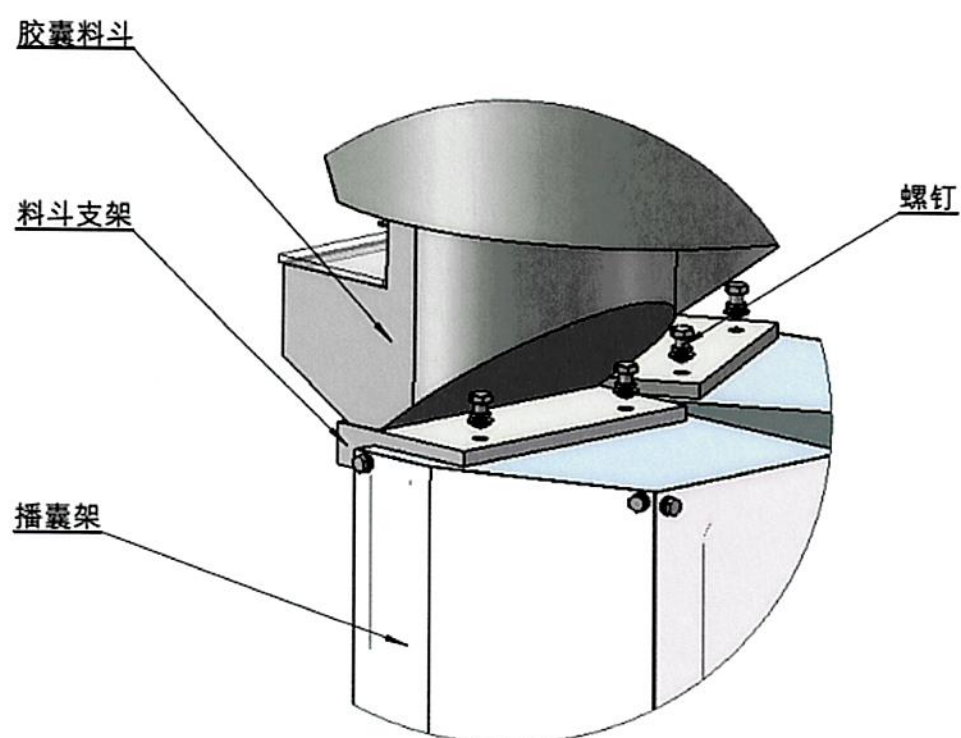
Filling guide pillar

#### 1.6.4 更换拆卸播囊理序装置

#### 1.6.4 Replacement and disassembly of the capsule feeding device

- a、 松开胶囊斗后方的 4 枚螺钉，取下螺钉和胶囊料斗。

Loosen the 4 screws at the back of the capsule hopper and remove the screws and the capsule material hopper.



胶囊料斗

Capsule hopper

螺钉

screw

料斗支架

Hopper bracket

播囊架

Seed bag holder

b、 打开传动部分侧门板，拿出附件工具中的手轮，套入电机输入轴后，用手转动主电机，使胶囊管部分上升至最高点。

b. Open the side door of the transmission section, take out the handwheel



from the accessory tools, fit it into the motor input shaft, and manually rotate the main motor to raise the capsule tube to the highest point.

c、 松开播囊管组件与滑板连接的螺钉，取下播囊管组件；松开胶囊梳上的螺钉，取下胶囊梳；松开送囊头与调节支板连接的螺钉，取下送囊头。

c. Loosen the screws connecting the bladder tube assembly to the slider and remove the bladder tube assembly; loosen the screws on the bladder comb and remove the bladder comb; loosen the screws connecting the bag head to the adjusting support plate and remove the bag head.

（注意：安装、换取时应尽量避免模具及零部件磕碰）

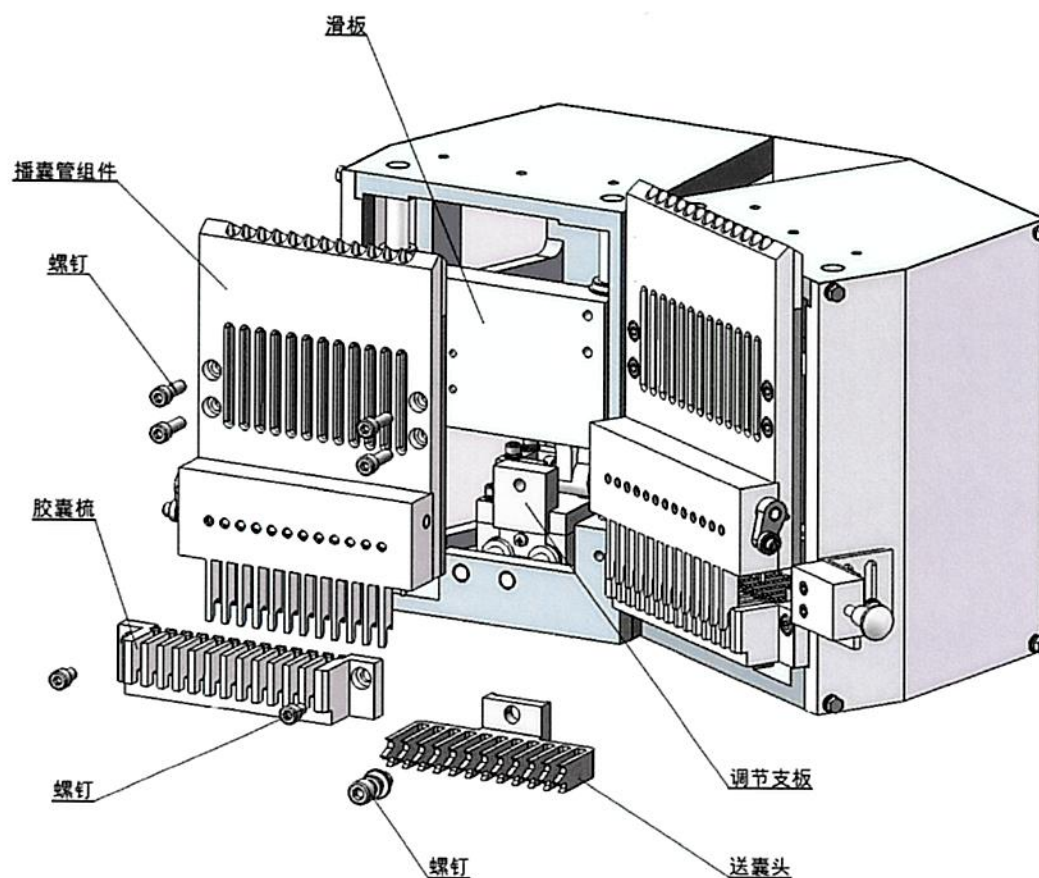
(Note: When installing or replacing, try to avoid bumping the mold and components)

d、 将准备替换的模具及零部件（播囊管组件、胶囊梳、送囊头）按序重新安装，取附件工具中相对应的校棒依照校对上下模具的方式进行校正好中心及间隙后，固紧相关螺钉并安装胶囊料斗。

d. Reinstall the replacement mold and components (bladder tube assembly, bladder comb, bag head) in sequence. Use the calibration rod in the accessory tools to calibrate the upper and lower molds in the way of aligning the center and gap, tighten the relevant screws, and install the bladder material hopper.

（注意：该机型采用两组播囊理序装置同时工作，但左右两组模具模孔数量不等，不可更换安装）

(Note: This machine model uses two sets of bladder sorting devices working simultaneously, but the number of mold holes in the left and right sets of molds is different and cannot be interchanged)



滑板

Slider

播囊管组件

Capsule tube assembly

螺钉

screw

胶囊梳



Capsule comb

**调节支板**

Adjusting support plate

**螺钉**

screw

**螺钉**

screw

**送囊头**

Feeding nozzle head

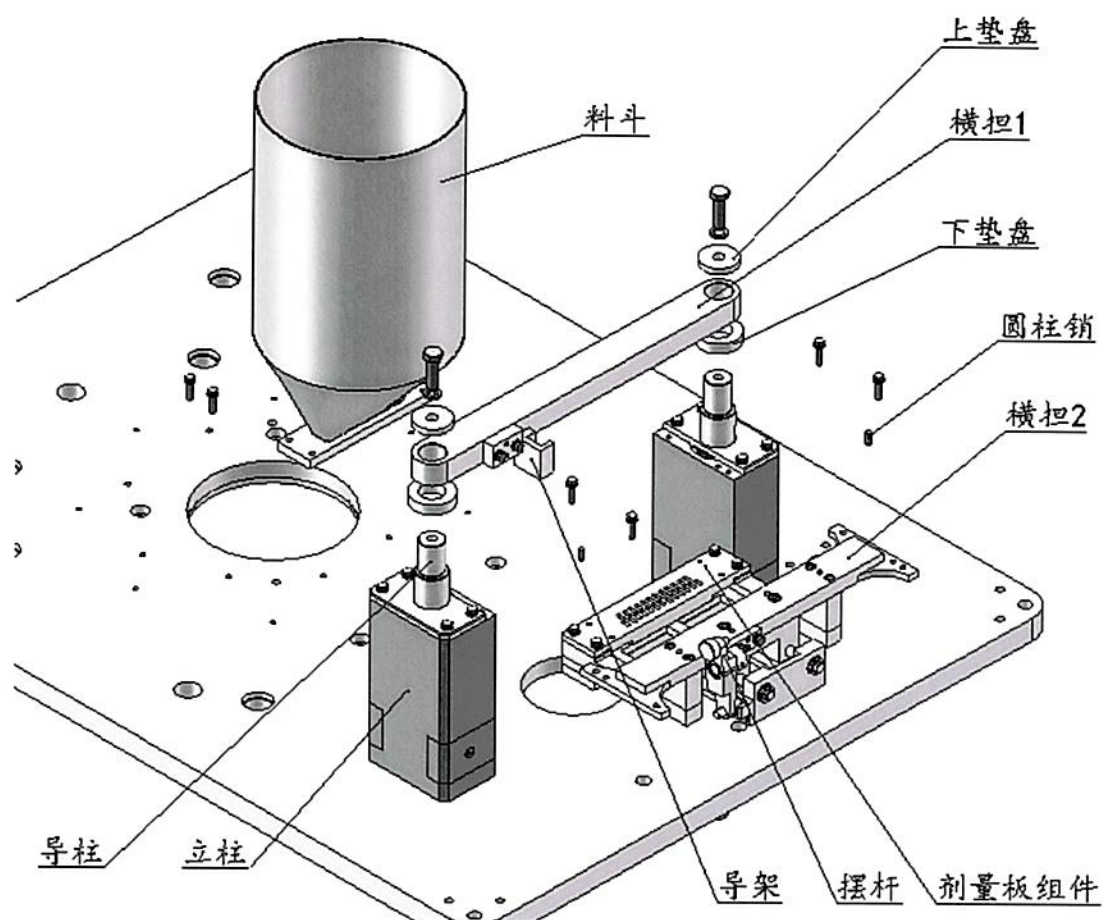
各模具及零部件更换完毕后，用手轮手动转动主电机输入轴，观察台面上所有零部件运动情况，待工位部件连续转 3—5 周后，皆无异常后，才能正常开机。（注意：开机后转速因由慢到快）

After replacing all molds and components, manually rotate the main motor input shaft with the handwheel, observe the movement of all components on the table, and only start the machine after the components at the workstation have rotated continuously for 3-5 cycles without any abnormalities. (Note: The speed increases gradually after the machine is turned on)

### 1.6.5 粉料填充装置更换为微丸灌装装置

#### 1.6.5 Powder Filling Device Replaced with Microsphere Filling Device

- a. 依照 1.6.3 拆卸粉末剂量盘的方式，将填充部分进行拆卸。
  - a. Disassemble the filling part according to the method of disassembling the powder dosing disc in 1.6.3.
- b. 松掉底盘与面板连接的螺钉，将余下的底盘调节装置继续进行拆卸。
  - b. Loosen the screws connecting the base plate to the panel, and continue to disassemble the remaining base plate adjustment device.
- c. 将横担 2 放置立柱上，对应销孔插入圆柱销，螺栓紧固。
  - c. Place crossbar 2 on the column, insert the corresponding pin hole into the cylindrical pin, and tighten the bolt.
- d. 与导柱同心依次放入下垫盘、横担 1、上垫盘并用螺栓紧固。
  - d. Place the lower pad, crossbar 1, and upper pad concentrically with the guide post and fasten them with bolts.
- e. 将料斗放置于剂量板组件上，对应螺栓孔紧固。
  - e. Place the hopper on the dosing plate assembly and fasten it to the corresponding bolt holes.



导柱

Guide pillar

立柱

mast

导架

Guide frame

摆杆

Swing arm

剂量板组件

Dosing plate assembly

上垫盘

Upper pad

料斗

Hopper

横担 1

Crossarm 1

下垫盘

Lower pad

圆柱销

Cylindrical pin

横担 2

Crossarm 2

,

## 四、机器相关图表

### IV. Machine-related charts

#### 4.1、润滑系统表

#### 4.1 Lubrication System Table

| 润滑点<br>Lubrication<br>point       | 应用类型<br>Application<br>type | 制造商名<br>Manufacturer<br>name  | 油型号<br>Oil type                |
|-----------------------------------|-----------------------------|-------------------------------|--------------------------------|
| 链条<br>Chain                       |                             | 润滑油公司<br>Lubricant<br>company | N4B GB443-84<br>N4B GB443-84   |
| 滚柱轴承<br>Tapered roller<br>bearing |                             | 润滑油公司<br>Lubricant<br>company | N4B GB443-84<br>N4B GB443-84   |
| 导向部件<br>Guide<br>component        |                             | 润滑油公司<br>Lubricant<br>company | N4B GB443-84<br>N4B GB443-84   |
| 凸轮<br>CAM                         |                             | 润滑油公司<br>Lubricant<br>company | ZL2 SY1412-75<br>ZL2 SY1412-75 |
| 配电箱                               |                             | 润滑油公司                         | ZLD SY1412-75                  |

|                              |  |                               |                                |
|------------------------------|--|-------------------------------|--------------------------------|
| Power<br>distribution<br>box |  | Lubricant<br>company          | ZLD SY1412-75                  |
| 减速机<br>reducer               |  | 润滑油公司<br>Lubricant<br>company | ZLD SY1412-75<br>ZLD SY1412-75 |



## 4.2、电控箱元件清单

### 4.2 Electrical Control Box Component List

| 称呼<br>Appellation                                         | 规格型号<br>Specifications and<br>models                                | 数量<br>Quantity | 生产厂家<br>Manufacturer       |
|-----------------------------------------------------------|---------------------------------------------------------------------|----------------|----------------------------|
| 触摸屏<br>touch panel                                        | SMART 1000 IEV3 10<br>寸 彩色<br>SMART1000IEV310<br>Inch Color Display | 1              | SIEMENS 西门                 |
| PLC 可编程控制<br>器<br>PLC<br>programmable logic<br>controller | CPU222 8 进 6 出                                                      | 1              | SIEMENS 西门                 |
| 扩展模块<br>Expansion<br>module                               | Em232                                                               | 1              | SIEMENS 西门<br>子<br>SIEMENS |
| 开关电源<br>Switching<br>Power Supply                         | 24VDC 35W S-35-24<br>24VDC 35W S-35-24                              | 1              | 台湾明伟<br>Taiwan Ming<br>Wei |



|                             |                                  |   |                                               |
|-----------------------------|----------------------------------|---|-----------------------------------------------|
| 交流接触器                       | 3TB41                            | 1 | SIEMENS 西门子<br>SIEMENS                        |
|                             | 3TB41                            | 2 |                                               |
|                             | 3TB43                            |   |                                               |
|                             | 3TB43                            |   |                                               |
| 中间继电器<br>Intermediate relay | MY4NJ<br>MY4NJ                   | 6 | 日本欧姆龙<br>Japan Omron                          |
| 空气开关                        | DZ47-60-C32                      | 1 | CHNT 公司                                       |
|                             | DZ47-60-C32                      | 1 |                                               |
|                             | DZ47-60-C5                       |   |                                               |
|                             | DZ47-60-C5                       |   |                                               |
| 变频调速器                       | Ev1000 系列<br>Ev1000 series       | 1 | 美国艾默生电气公司 (安圣)<br>Emerson Electric Co. (ASCO) |
| 主机电机<br>Main machine motor  | R57DRS112M4/2W<br>R57DRS112M4/2W | 1 | 德国 SEW 公司<br>SEW company from Germany         |
| 真空泵<br>vacuum pump          | SK 系列<br>SK series               | 1 | 浙江新环<br>Zhejiang Xinhuan                      |



|                                                                 |                                                                                                |                      |                                           |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------|
| <b>吸尘器</b><br>Vacuum<br>cleaner                                 | <b>EDC 系列</b><br>EDC series                                                                    | <b>1</b>             | <b>本厂生产</b><br>Produced by our<br>factory |
| <b>高精度</b><br>High<br>Resolution<br><b>凸轮分箱</b><br>Cam splitter | <b>RU180DF-12-150</b><br><b>RU180DF-12-150</b><br><b>RU140DF-6-120</b><br><b>RU140DF-6-120</b> | <b>1</b><br><b>1</b> | <b>台湾翔笙</b>                               |
| <b>料位检测器</b><br>Level<br>Detector                               | <b>E2K-C25ME1</b><br><b>E2K-C25ME1</b>                                                         | <b>1</b>             | <b>欧姆龙</b><br>Omron                       |
| <b>分断器</b><br>Circuit<br>Breaker                                | <b>11E 3VE1</b><br><b>11E 3VE1</b>                                                             | <b>3</b>             | <b>SIEMENS 西门子</b><br><b>SIEMENS</b>      |

## 4.3、轴承目录

## 4.3, bearing catalog

## 工位部分

## Workstation Section

| 名称<br>Name                                     | 规格<br>Specifications             | 数量<br>Quantity |
|------------------------------------------------|----------------------------------|----------------|
| 直线滚动轴承<br>Linear rolling bearings              | zxt 16 26 36<br>zxt 16 26 36     | 96             |
| 深沟球轴承 6302<br>Deep groove ball<br>bearing 6302 | d=15 D=42 B=13<br>d=15 D=42 B=13 | 12             |
| 连轴轴承 CF-6<br>Cam follower bearing              | D=6 D=16 B=11<br>D=6 D=16 B=11   | 12             |

CF-6

## 播囊部分

## Capsule Section

| 名称<br>Name                                     | 规格<br>Specifications             | 数量<br>Quantity |
|------------------------------------------------|----------------------------------|----------------|
| 直线滚动轴承<br>Linear rolling bearings              | zxt 16 26 36<br>zxt 16 26 36     | 8              |
| 直线滚动轴承<br>Linear rolling bearings              | zxt 12 22 32<br>zxt 12 22 32     | 8              |
| 深沟球轴承 625<br>Deep groove ball<br>bearing 625   | d=5 D=16 B=5<br>d=5 D=16 B=5     | 2              |
| 深沟球轴承 626<br>Deep groove ball<br>bearing 626   | d=6 D=19 B=6<br>d=6 D=19 B=6     | 2              |
| 深沟球轴承 608<br>Deep groove ball<br>bearing 608   | d=8 D=22 B=7<br>d=8 D=22 B=7     | 4              |
| 深沟球轴承 6004<br>Deep groove ball<br>bearing 6004 | d=20 D=42 B=12<br>d=20 D=42 B=12 | 4              |

## 充填立柱部分

### Filled column section

| 名称<br>Name                        | 规格<br>Specifications           | 数量<br>Quantity |
|-----------------------------------|--------------------------------|----------------|
| 直线滚动轴承<br>Linear rolling bearings | zxt 50 80 100<br>zxt 50 80 100 | 4              |

## 传动部分

### Transmission Part

| 名称<br>Name                                       | 规格<br>Specifications               | 数量<br>Quantity |
|--------------------------------------------------|------------------------------------|----------------|
| 直线滚动轴承<br>Linear rolling bearings                | zxt 25 40 58<br>zxt 25 40 58       | 16             |
| 直线滚动轴承<br>Linear rolling bearings                | zxt 12 22 32<br>zxt 12 22 32       | 3              |
| 调心球轴承 1309<br>Self-aligning ball<br>bearing 1309 | d=50 D=100 B=25<br>d=50 D=100 B=25 | 2              |
| 调心球轴承 1507                                       | d=35 D=72 B=23                     | 1              |



|                                                      |                                  |    |
|------------------------------------------------------|----------------------------------|----|
| Self-aligning ball bearing 1507                      | d=35 D=72 B=23                   |    |
| 连轴轴承 CF-18<br>Continuous shaft bearing CF-18         | d=18 D=40 B=20<br>d=18 D=40 B=20 | 7  |
| 深沟球轴 62200-2RS<br>Deep groove ball bearing 62200-2RS | d=10 D=30 B=14<br>d=10 D=30 B=14 | 8  |
| 滚针轴承 NA4904<br>Needle roller bearing NA4904          | d=20 D=37 c=17<br>d=20 D=37 c=17 | 1  |
| 滚针轴承 NA4905<br>Needle roller bearing NA4905          | d=25 D=42 c=17<br>d=25 D=42 c=17 | 10 |
| 滚针轴承 NA4906<br>Needle roller bearing NA4906          | d=30 D=47 c=17<br>d=30 D=47 c=17 | 2  |